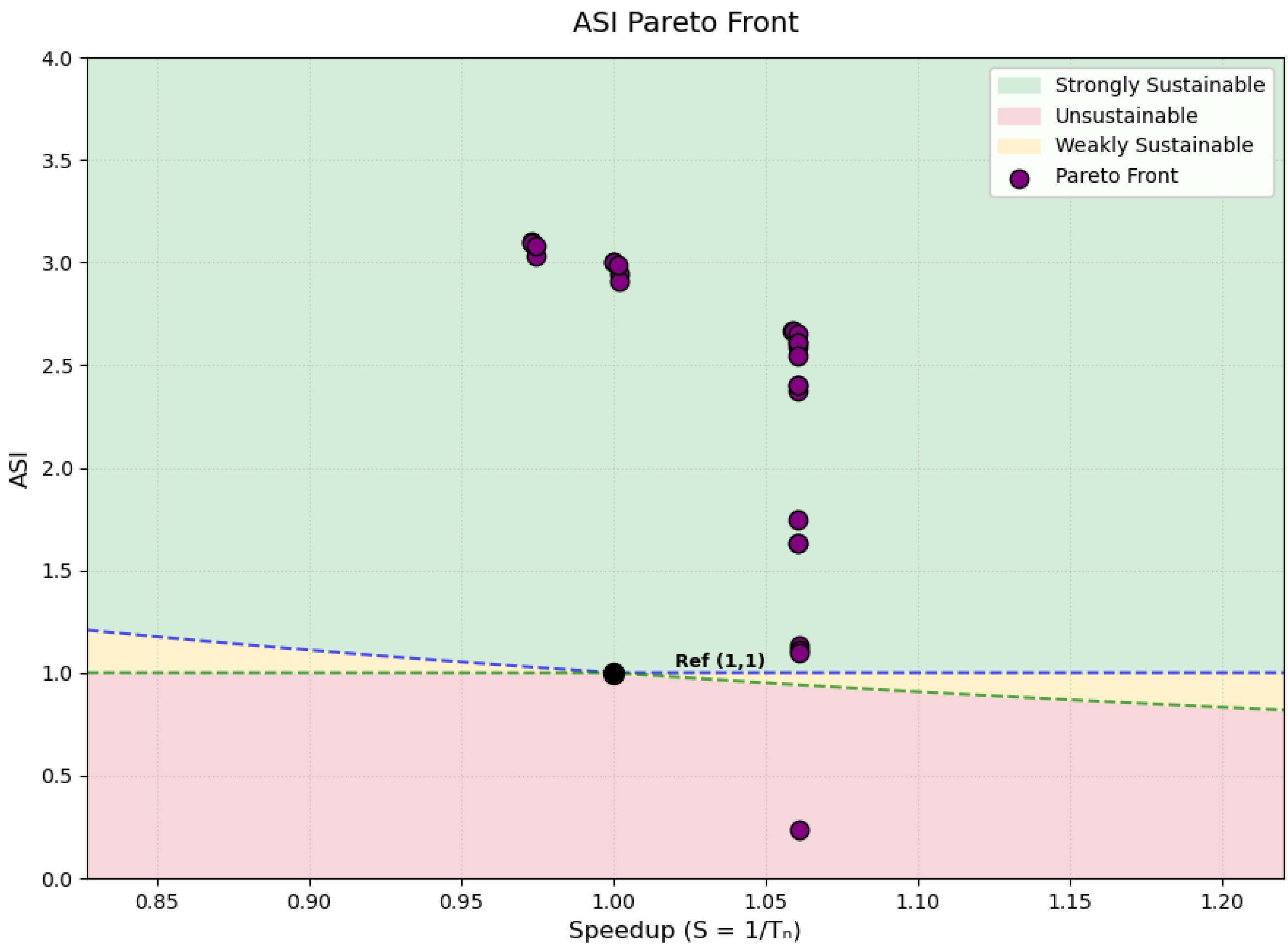
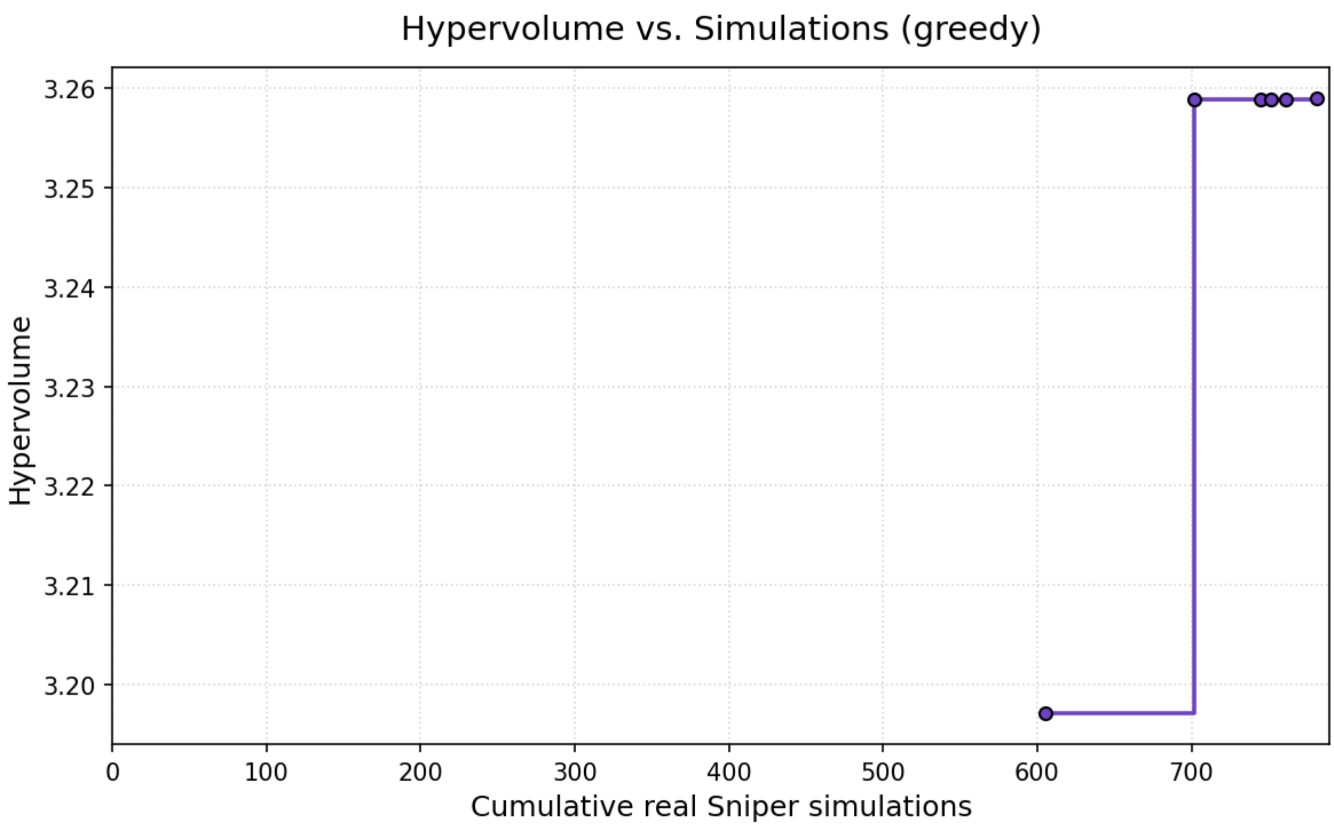
ML2 benchmark:

ML2: L2 resident (depending on LFSR settings) linked list traversal

Greedy





Result:

Total sniper runs: 781 Final hypervolume: 3.2594

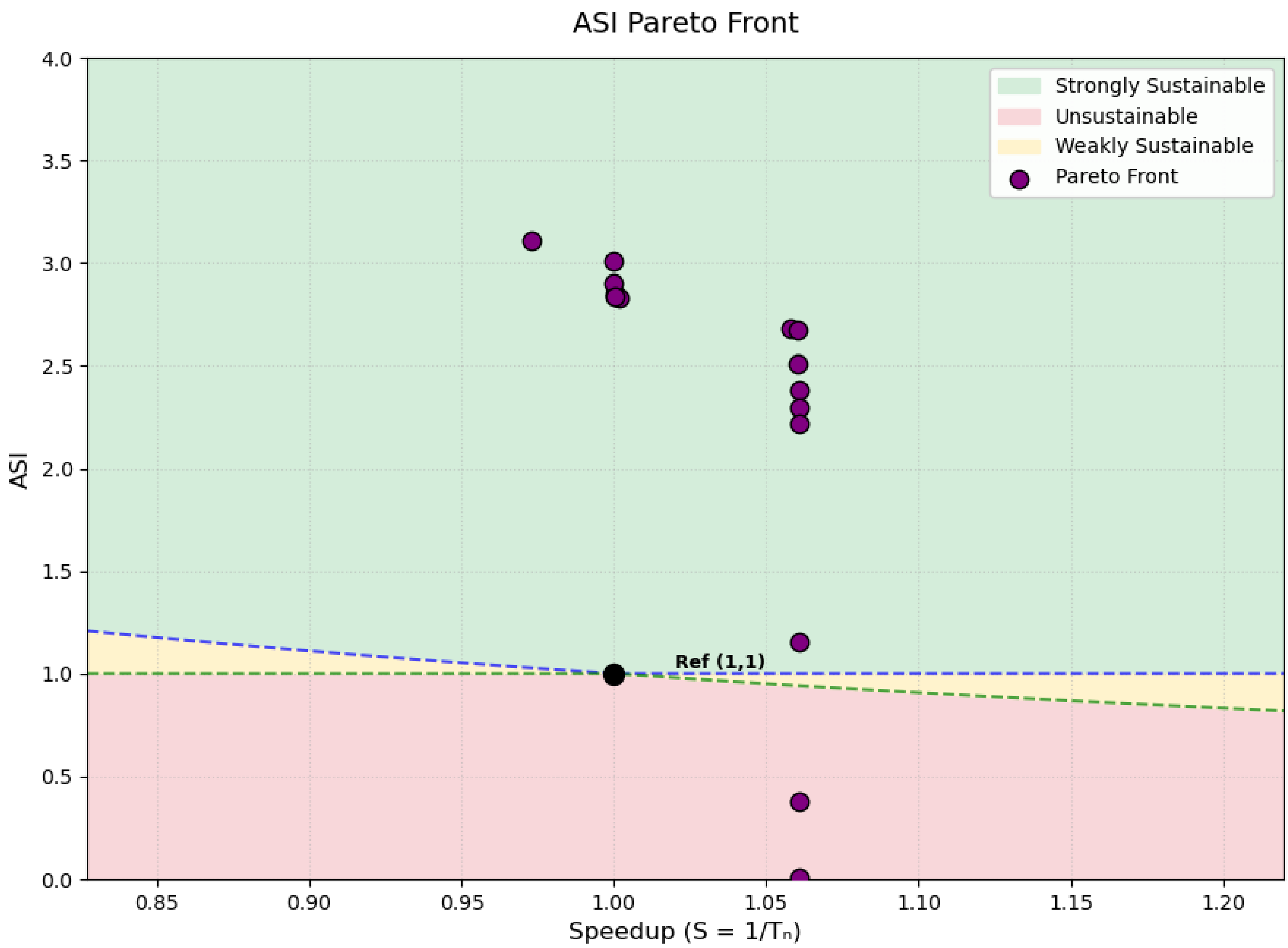
Front:

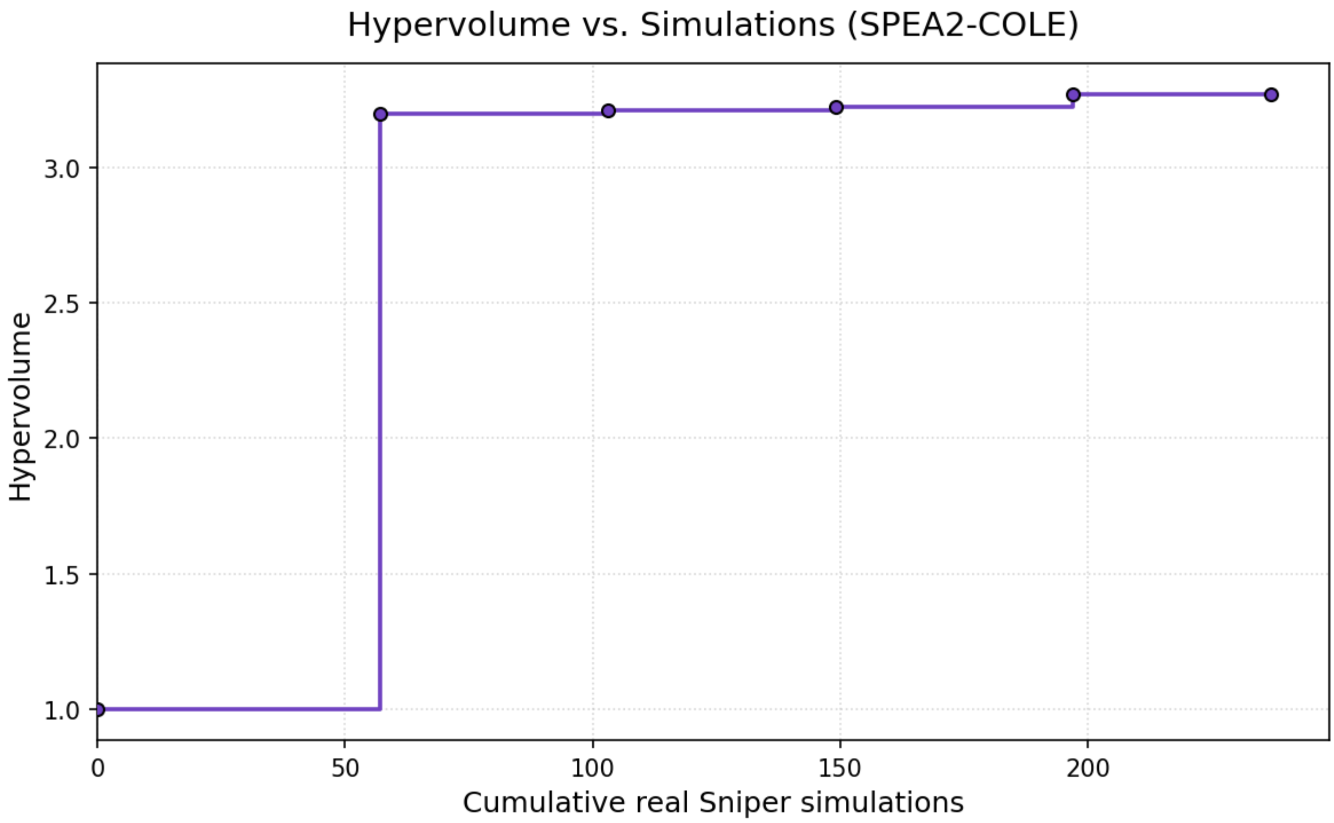
bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
a53		4	64				512		2048	4			1024	0.2323	1.0610	67.94	141.87	Unsustainable
pentium_m		4	64	8	16		512		1024		8	2	1024	1.1008	1.0610	49.33	34.59	Strongly Sust.
pentium_m		4	64	8			512	4	1024		2	2	1024	1.1154	1.0609	47.61	34.56	Strongly Sust.
pentium_m		4	64				512		1024			2	1024	1.1342	1.0608	47.96	33.90	Strongly Sust.
a53			64				512		2048		2			1.6355	1.0608	55.79	22.20	Strongly Sust.
a53	512		64				512		2048		2			1.6355	1.0608	55.79	22.20	Strongly Sust.
a53			64				512		2048				16	1.7450	1.0607	55.19	20.90	Strongly Sust.
a53	512		64				512		1024			2		2.3749	1.0607	47.80	16.21	Strongly Sust.
pentium_m		4	64	8			512		1024			2	64	2.4038	1.0607	45.04	16.33	Strongly Sust.
a53	512	4	64	8			512		1024			2	64	2.4038	1.0606	45.04	16.33	Strongly Sust.
a53	512	4	64				512		1024			2	64	2.5447	1.0606	43.32	15.61	Strongly Sust.
a53		4	64				512	8	1024			2	64	2.5897	1.0606	41.24	15.56	Strongly Sust.
a53	512	4	64				512		1024			2	16	2.6083	1.0606	43.17	15.25	Strongly Sust.
a53		4	64				512		1024			2	16	2.6083	1.0605	43.17	15.25	Strongly Sust.
a53		4	64				512	8	1024			2	16	2.6546	1.0605	41.09	15.19	Strongly Sust.
a53		4					512	8	1024			2	16	2.6666	1.0589	40.92	15.14	Strongly Sust.
a53		4					512	4	1024			2	16	2.6682	1.0589	40.92	15.13	Strongly Sust.
a53		4	64			4			1024			2	32	2.9076	1.0019	36.49	14.31	Strongly Sust.
a53		4	64			4			1024			2	16	2.9477	1.0018	36.40	14.12	Strongly Sust.
a53		4	64					8	1024			2	16	2.9861	1.0013	34.86	14.08	Strongly Sust.
a53		4						8	1024			2	16	3.0002	0.9999	34.69	14.03	Strongly Sust.
a53		4						4	1024			2	16	3.0022	0.9999	34.69	14.02	Strongly Sust.
a53		4	64			4	128		1024			2	16	3.0325	0.9746	35.43	13.81	Strongly Sust.
a53		4	64				128	8	1024			2	16	3.0823	0.9744	33.37	13.77	Strongly Sust.
a53		4					128	8	1024			2	16	3.0970	0.9732	33.20	13.72	Strongly Sust.
a53		4					128	4	1024			2	16	3.0991	0.9731	33.20	13.71	Strongly Sust.

SPEA2 - COLE

hyperparameter settings 1

Hyperparameter settings: Patience = 1 (amount of iterations to wait before stopping after no improvement), max_iterations: int = 10. (max iter never reached)





Result:

Total sniper runs: 237 Final hypervolume: 3.2693

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
a53	1024	4	64	8	16	8	512	16	1024	3	2	8	1024	0.0070	1.0608	157.56	1188.00	Unsustainable
pentium_m		8	64	4	16	8	512	16	1024		2	4	512	0.3748	1.0608	57.72	95.43	Unsustainable
pentium_m		4	64	4	16	8	512	4	1024		2	2	1024	1.1558	1.0608	45.64	33.82	Strongly Sust.
pentium_m		4	64	8	64	4	512	8	2048		4	2	256	2.2207	1.0607	48.56	17.24	Strongly Sust.
tage		4	64	8	16	8	512	4	2048		10	2	16	2.2940	1.0607	49.42	16.59	Strongly Sust.
pentium_m		4	64	4	32	8	512	16	2048		4	2	32	2.3800	1.0606	48.12	16.14	Strongly Sust.
pentium_m		4	64	8	64	8	512	4	1024		10	2	16	2.5075	1.0605	43.13	15.86	Strongly Sust.
tage		4	64	4	16	4	512	4	1024		8	2	16	2.6748	1.0605	40.76	15.11	Strongly Sust.
tage		4	16	4	32	8	512	4	1024		4	2	16	2.6778	1.0581	40.73	15.10	Strongly Sust.
pentium_m		4	64	8	64	4	256	8	1024		2	2	16	2.8286	1.0018	36.35	14.72	Strongly Sust.
pentium_m		4	32	8	64	4	256	8	1024		10	2	16	2.8417	1.0004	36.18	14.67	Strongly Sust.
pentium_m		4	32	8	64	4	256	8	1024		10	2	16	2.8417	1.0004	36.18	14.67	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	1024		10	2	16	2.8550	1.0003	35.52	14.66	Strongly Sust.
pentium_m		4	32	4	16	8	256	4	1024		10	2	128	2.9034	1.0000	34.66	14.50	Strongly Sust.
pentium_m		4	32	4	16	8	256	4	1024		2	2	16	3.0118	0.9999	34.45	14.00	Strongly Sust.
pentium_m		4	32	4	16	8	128	4	1024		2	2	16	3.1091	0.9731	32.96	13.69	Strongly Sust.

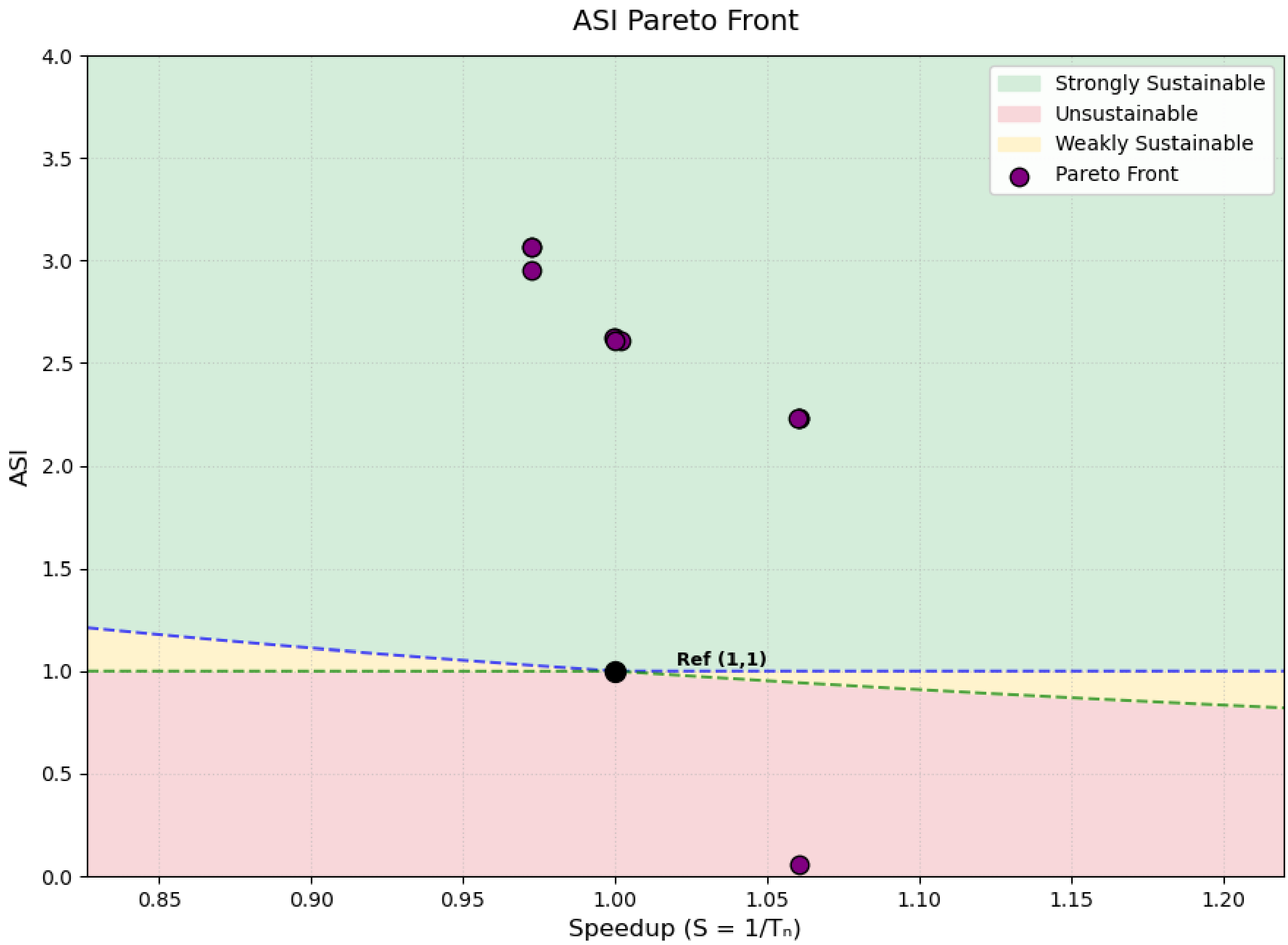
Max-value Entropy Search for Multi-objective Bayesian Optimization (MESMO)

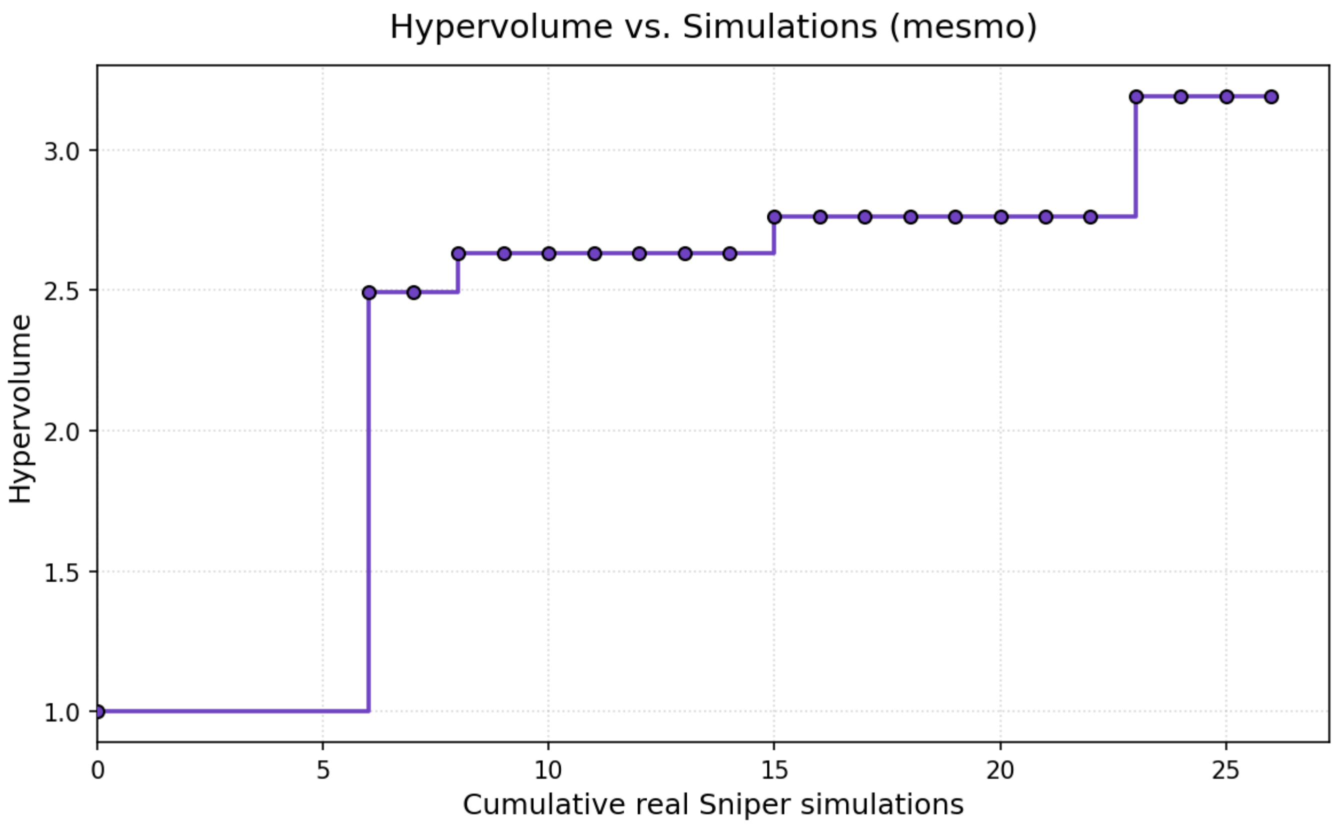
source: https://arxiv.org/pdf/2110.06980

hyperparameter settings 1

Hyperparameter settings: 400 candidate samples (400 candidate configurations are considered per iteration; each is scored by the information gained, averaged over a default of 10 Monte Carlo posterior-function samples (amount is hyperparameter, default is 10), and the top-scoring one (also hyperparameter, default is 1) is evaluated.) are used and 7 initial configurations (including baseline, always the case!) have been run.

20 iterations





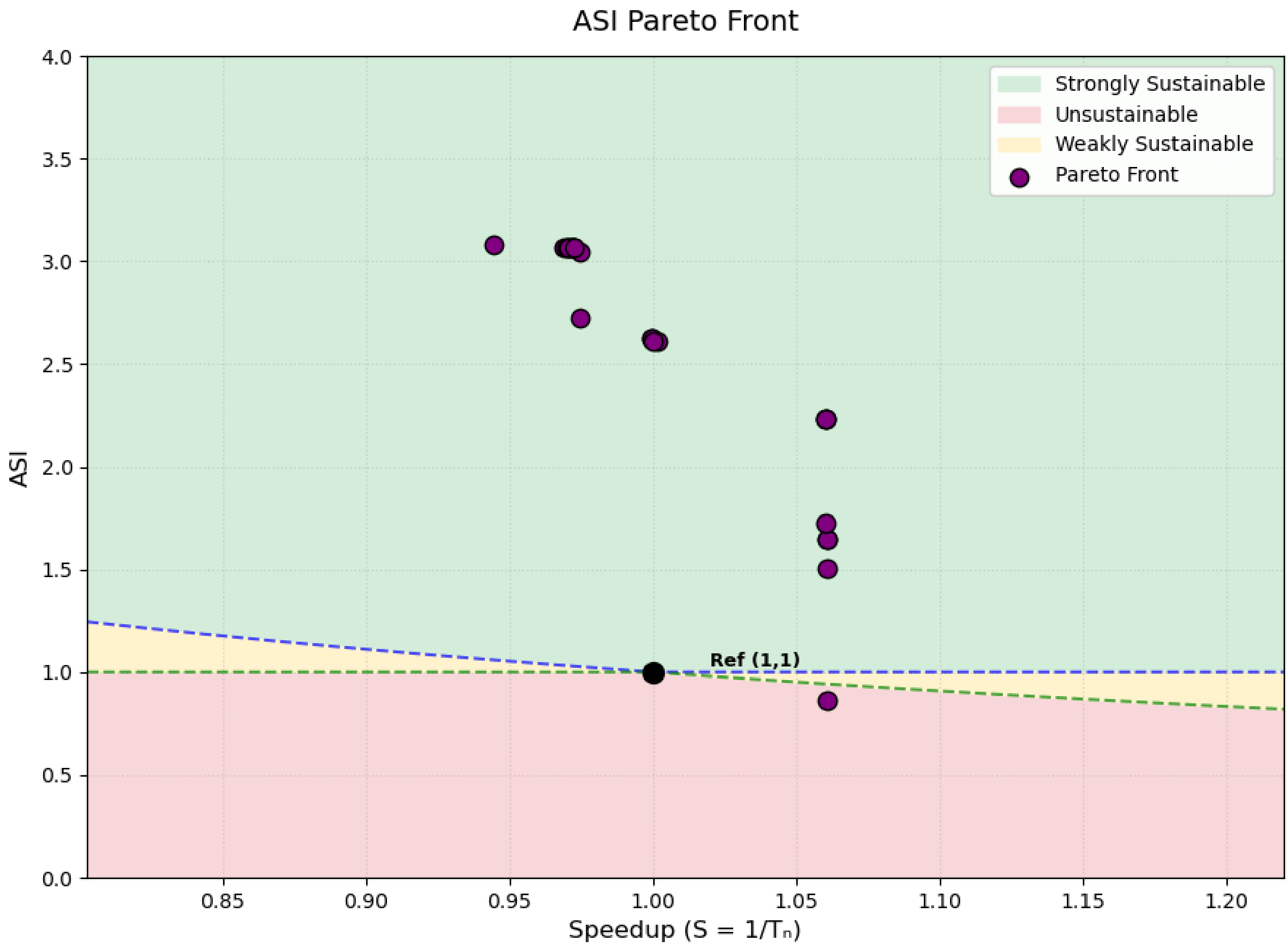
Result:

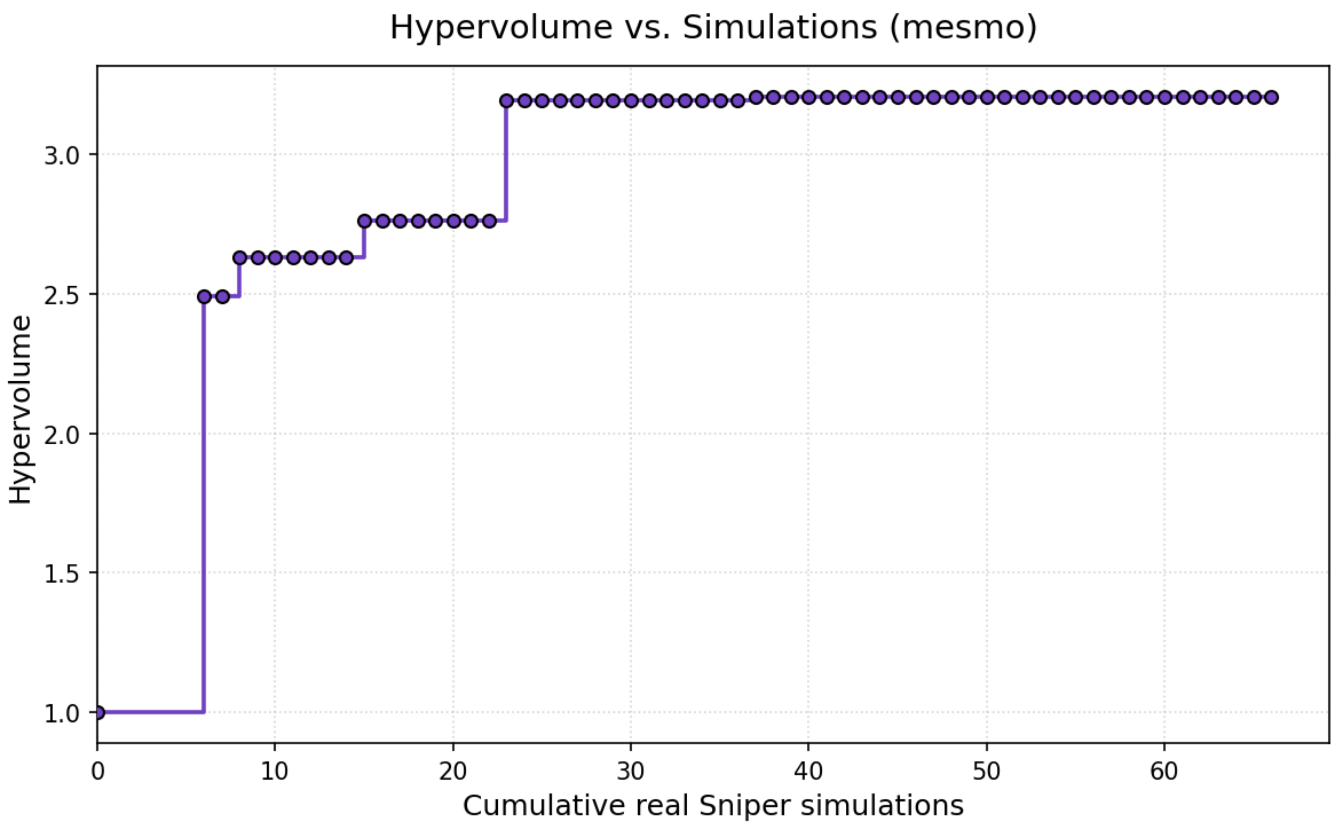
Total sniper runs: 26 Final hypervolume: 3.1908

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m		8	64	8	16	8	512	16	1024				10	8	256	0.0537	1.0608	83.61	533.35	Unsustainable
a53	512	8	64	8	16	8	512	16	1024	2			10	2	256	2.2291	1.0606	49.29	17.09	Strongly Sust.
pentium_m		8	64	8	16	8	512	16	1024				10	2	256	2.2292	1.0605	49.29	17.08	Strongly Sust.
nn		8	64	8	16	8	512	16	1024		16	0.005	10	2	256	2.2293	1.0602	49.29	17.08	Strongly Sust.
a53	512	4	64	4	16	4	256	4	2048	4			2	2	256	2.6064	1.0018	41.36	15.45	Strongly Sust.
nn		4	64	4	16	4	256	4	2048		16	0.0005	2	2	256	2.6071	1.0001	41.36	15.44	Strongly Sust.
pentium_m		4	16	4	16	4	256	4	2048				2	2	256	2.6273	0.9999	40.99	15.36	Strongly Sust.
a53	512	4	16	4	16	4	256	4	2048	4			2	2	256	2.6273	0.9997	40.99	15.36	Strongly Sust.
a53	512	4	16	4	32	8	128	4	1024	4			2	2	256	2.9545	0.9724	33.33	14.37	Strongly Sust.
a53	512	4	16	4	32	8	128	4	1024	4			4	2	32	3.0672	0.9723	33.09	13.87	Strongly Sust.
a53	512	4	16	4	32	8	128	4	1024	4			2	2	32	3.0672	0.9723	33.09	13.87	Strongly Sust.

60 iterations





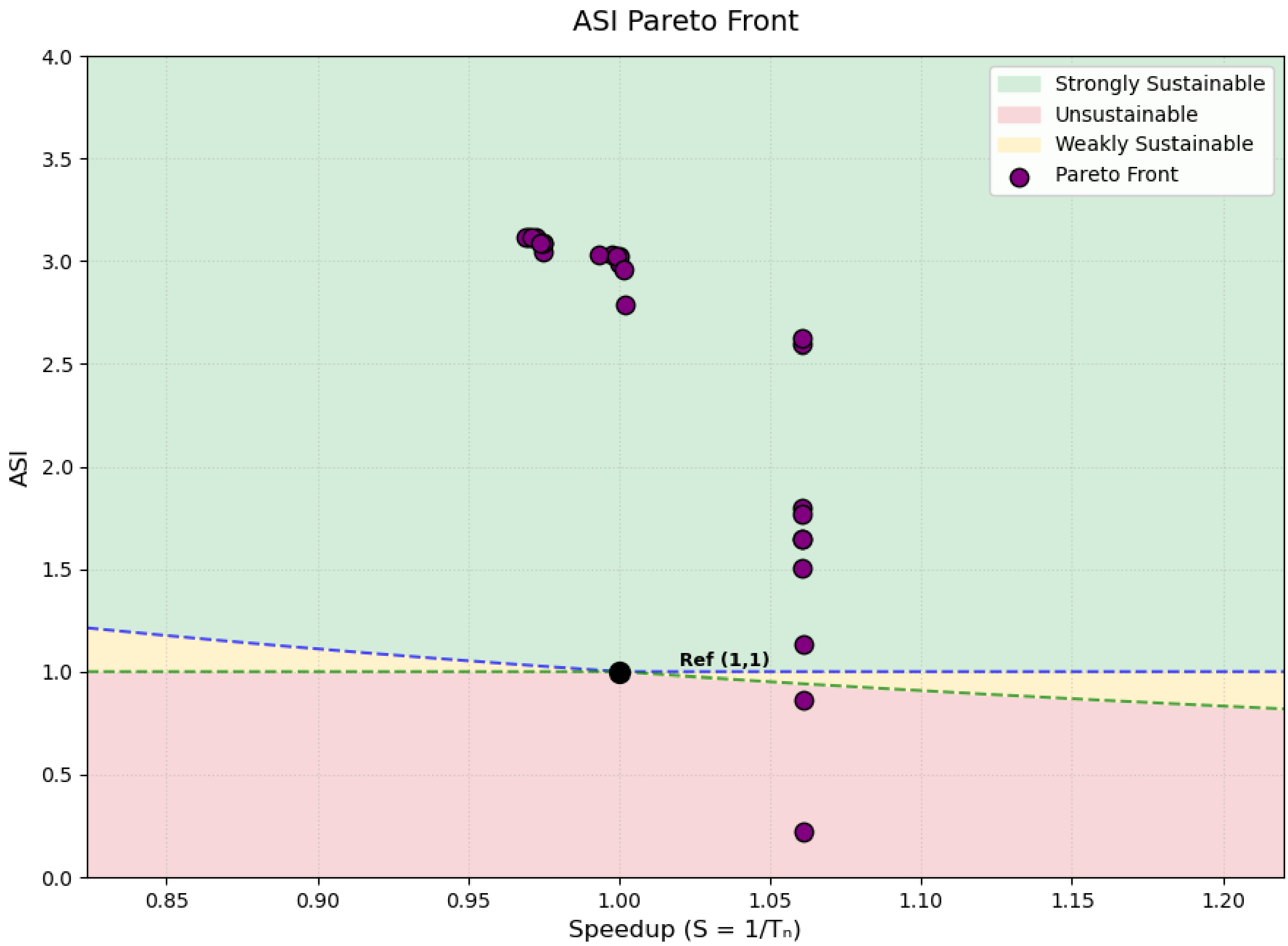
Result:

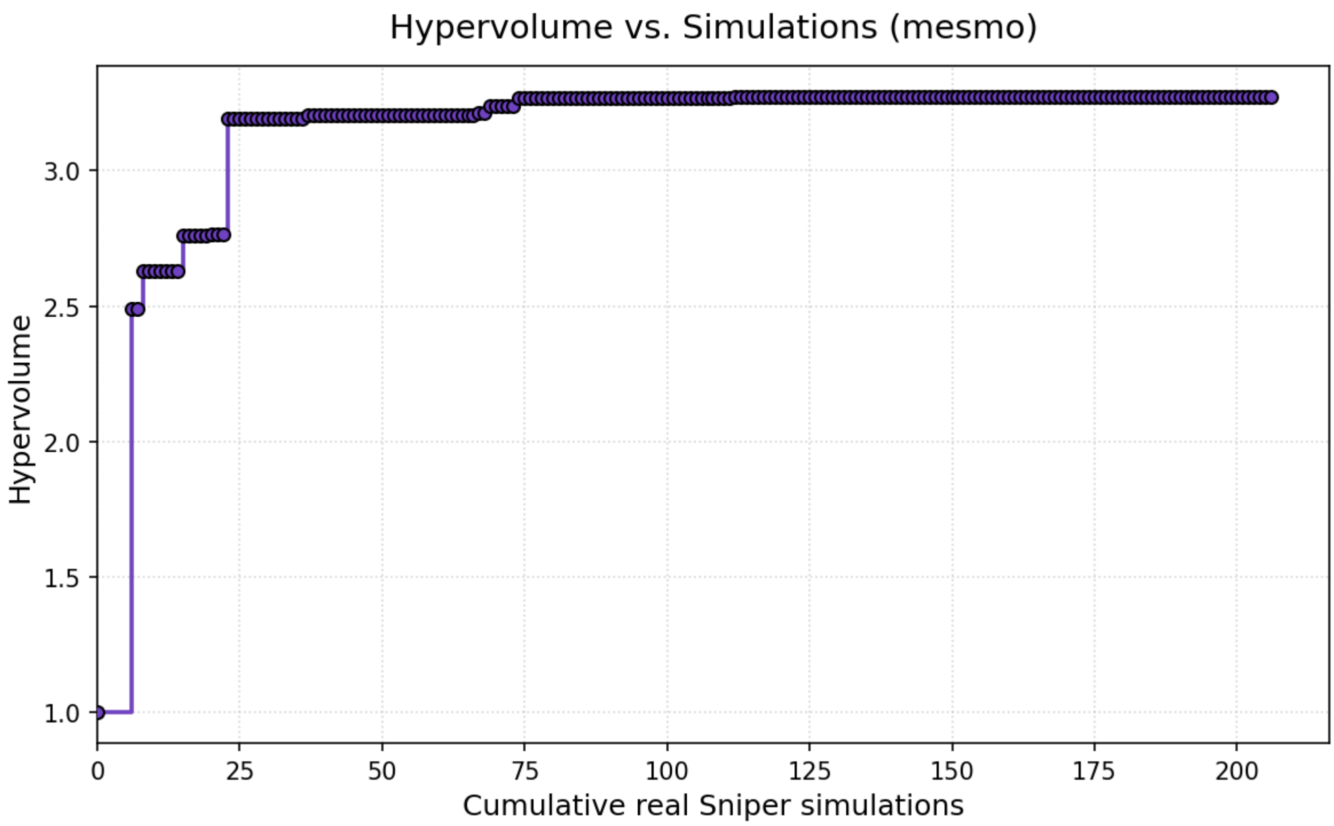
Total sniper runs: 66 Final hypervolume: 3.2056

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m		8	64	8	16	4	512	8	4096				10	2	1024	0.8626	1.0609	70.94	37.26	Unsustainable
pentium_m		8	64	8	16	8	512	16	1024				10	2	512	1.5034	1.0608	50.78	25.06	Strongly Sust.
pentium_m		4	64	8	64	8	512	16	2048				10	4	128	1.6488	1.0608	53.40	22.41	Strongly Sust.
pentium_m		4	64	8	64	8	512	16	2048				8	4	128	1.6488	1.0607	53.40	22.41	Strongly Sust.
a53	512	8	64	8	16	8	512	16	4096	2			10	2	256	1.7280	1.0606	66.19	19.35	Strongly Sust.
a53	512	8	64	8	16	8	512	16	1024	2			10	2	256	2.2291	1.0606	49.29	17.09	Strongly Sust.
pentium_m		8	64	8	16	8	512	16	1024				10	2	256	2.2292	1.0605	49.29	17.08	Strongly Sust.
nn		8	64	8	16	8	512	16	1024		16	0.005	10	2	256	2.2293	1.0602	49.29	17.08	Strongly Sust.
a53	512	4	64	4	16	4	256	4	2048	4			2	2	256	2.6064	1.0018	41.36	15.45	Strongly Sust.
nn		4	64	4	16	4	256	4	2048		16	0.0005	2	2	256	2.6071	1.0001	41.36	15.44	Strongly Sust.
pentium_m		4	16	4	16	4	256	4	2048				2	2	256	2.6273	0.9999	40.99	15.36	Strongly Sust.
a53	512	4	16	4	16	4	256	4	2048	4			2	2	256	2.6273	0.9997	40.99	15.36	Strongly Sust.
tage		4	64	4	32	4	128	16	2048				4	2	64	2.7216	0.9747	40.44	14.89	Strongly Sust.
nn		4	64	4	32	8	128	4	1024		64	0.0005	8	2	32	3.0414	0.9746	33.46	13.95	Strongly Sust.
a53	512	4	16	4	32	8	128	4	1024	3			8	2	32	3.0671	0.9725	33.09	13.87	Strongly Sust.
a53	512	4	16	4	32	8	128	4	1024	4			10	2	32	3.0671	0.9724	33.09	13.87	Strongly Sust.
tage		4	16	4	32	8	128	4	1024				2	2	32	3.0672	0.9723	33.09	13.87	Strongly Sust.
a53	512	4	16	4	32	8	128	4	1024	4			8	2	32	3.0672	0.9723	33.09	13.87	Strongly Sust.
nn		4	16	4	32	8	128	4	1024		16	0.0005	10	2	32	3.0675	0.9717	33.09	13.87	Strongly Sust.
nn		4	16	4	32	8	128	4	1024		64	0.005	2	2	32	3.0681	0.9706	33.09	13.86	Strongly Sust.
nn		4	16	4	32	8	128	4	1024		64	0.005	10	2	32	3.0682	0.9703	33.09	13.86	Strongly Sust.
nn		4	16	4	32	8	128	4	1024		32	0.0005	10	2	32	3.0684	0.9700	33.09	13.86	Strongly Sust.
nn		4	16	4	32	8	128	4	1024		32	0.0005	8	2	32	3.0688	0.9692	33.09	13.86	Strongly Sust.
nn		4	16	4	32	8	128	4	1024		64	0.0005	8	2	32	3.0814	0.9447	33.09	13.80	Strongly Sust.

200 iterations





Result:

Total sniper runs: 206 Final hypervolume: 3.2729

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m		8	64	4	64	8	512	4	2048				8	4	1024	0.2219	1.0609	72.96	142.35	Unsustainable
pentium_m		8	64	8	16	4	512	8	4096				10	2	1024	0.8626	1.0609	70.94	37.26	Unsustainable
pentium_m		4	64	4	16	8	512	16	1024				2	2	1024	1.1369	1.0609	47.72	33.88	Strongly Sust.
pentium_m		8	64	8	16	8	512	16	1024				10	2	512	1.5034	1.0608	50.78	25.06	Strongly Sust.
pentium_m		4	64	8	64	8	512	16	2048				10	4	128	1.6488	1.0608	53.40	22.41	Strongly Sust.
pentium_m		4	64	8	64	8	512	16	2048				8	4	128	1.6488	1.0607	53.40	22.41	Strongly Sust.
pentium_m		8	64	8	64	8	512	16	4096				8	2	32	1.7692	1.0606	66.61	18.84	Strongly Sust.
pentium_m		4	64	8	32	8	512	16	1024				8	4	64	1.7965	1.0606	48.05	21.39	Strongly Sust.
pentium_m		4	64	4	16	8	512	8	1024				10	2	64	2.5978	1.0606	41.00	15.54	Strongly Sust.
nn		4	64	4	32	8	512	4	1024		64	0.0005	8	2	32	2.6225	1.0606	41.18	15.37	Strongly Sust.
tage		8	64	4	32	4	256	8	1024				2	2	32	2.7856	1.0019	38.83	14.70	Strongly Sust.
a53	2048	4	64	4	64	8	256	8	1024	2			4	2	16	2.9603	1.0013	35.45	14.15	Strongly Sust.
nn		4	32	4	64	4	256	4	1024		16	0.005	2	2	16	2.9891	1.0002	34.74	14.08	Strongly Sust.
nn		4	32	4	16	4	256	4	1024		16	0.005	10	2	16	3.0249	1.0002	33.91	13.99	Strongly Sust.
nn		4	16	4	32	4	256	4	1024		16	0.005	10	2	16	3.0264	0.9993	33.96	13.97	Strongly Sust.
nn		4	16	4	32	4	256	4	1024		64	0.005	10	2	16	3.0272	0.9977	33.96	13.97	Strongly Sust.
nn		4	32	4	16	4	256	4	1024		16	0.0005	10	2	16	3.0284	0.9932	33.91	13.97	Strongly Sust.
a53	2048	4	64	4	32	4	128	4	1024	4			8	2	32	3.0444	0.9746	33.43	13.94	Strongly Sust.
a53	2048	4	64	4	32	4	128	4	1024	4			8	2	16	3.0873	0.9746	33.35	13.75	Strongly Sust.
a53	512	4	64	4	32	4	128	4	1024	4			8	2	16	3.0873	0.9746	33.35	13.75	Strongly Sust.
nn		4	64	4	32	4	128	4	1024		16	0.0005	8	2	16	3.0877	0.9738	33.35	13.75	Strongly Sust.
tage		4	16	4	32	4	128	4	1024				10	2	16	3.1136	0.9724	32.98	13.67	Strongly Sust.
nn		4	16	4	32	4	128	4	1024		16	0.001	10	2	16	3.1144	0.9709	32.98	13.67	Strongly Sust.
nn		4	16	4	32	4	128	4	1024		64	0.005	10	2	16	3.1149	0.9699	32.98	13.66	Strongly Sust.
nn		4	16	4	32	4	128	4	1024		64	0.001	10	2	16	3.1153	0.9692	32.98	13.66	Strongly Sust.

HV was already 3.2727 at iter 152 beofre that also quite slowe increase. Amount of points did increase with 10.

ML2 and CCl benchmarks:

ML2: L2 resident (depending on LFSR settings) linked list traversal

CCl: Impossible control with large basic blocks (potentially larger penalty)

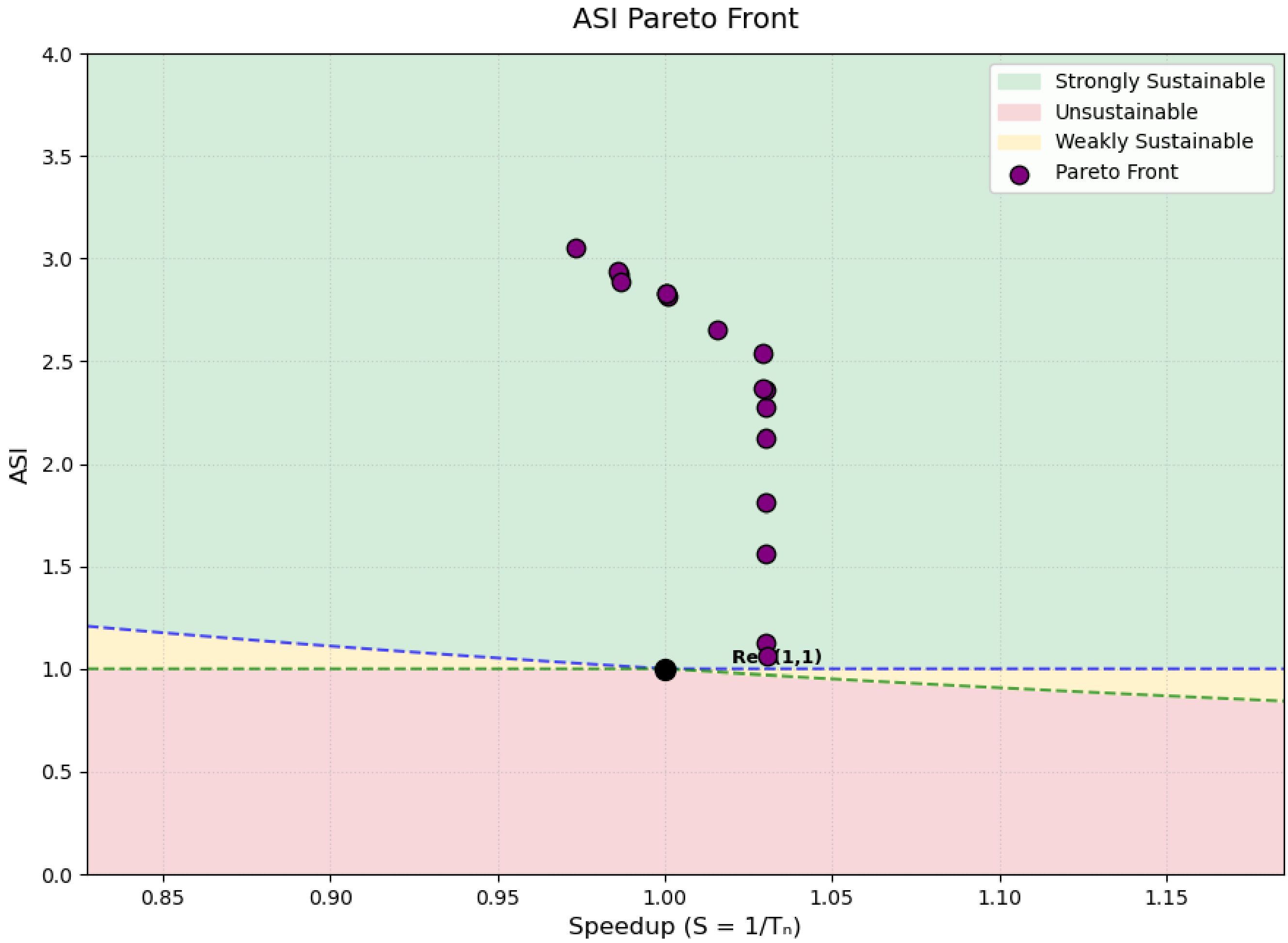
Greedy

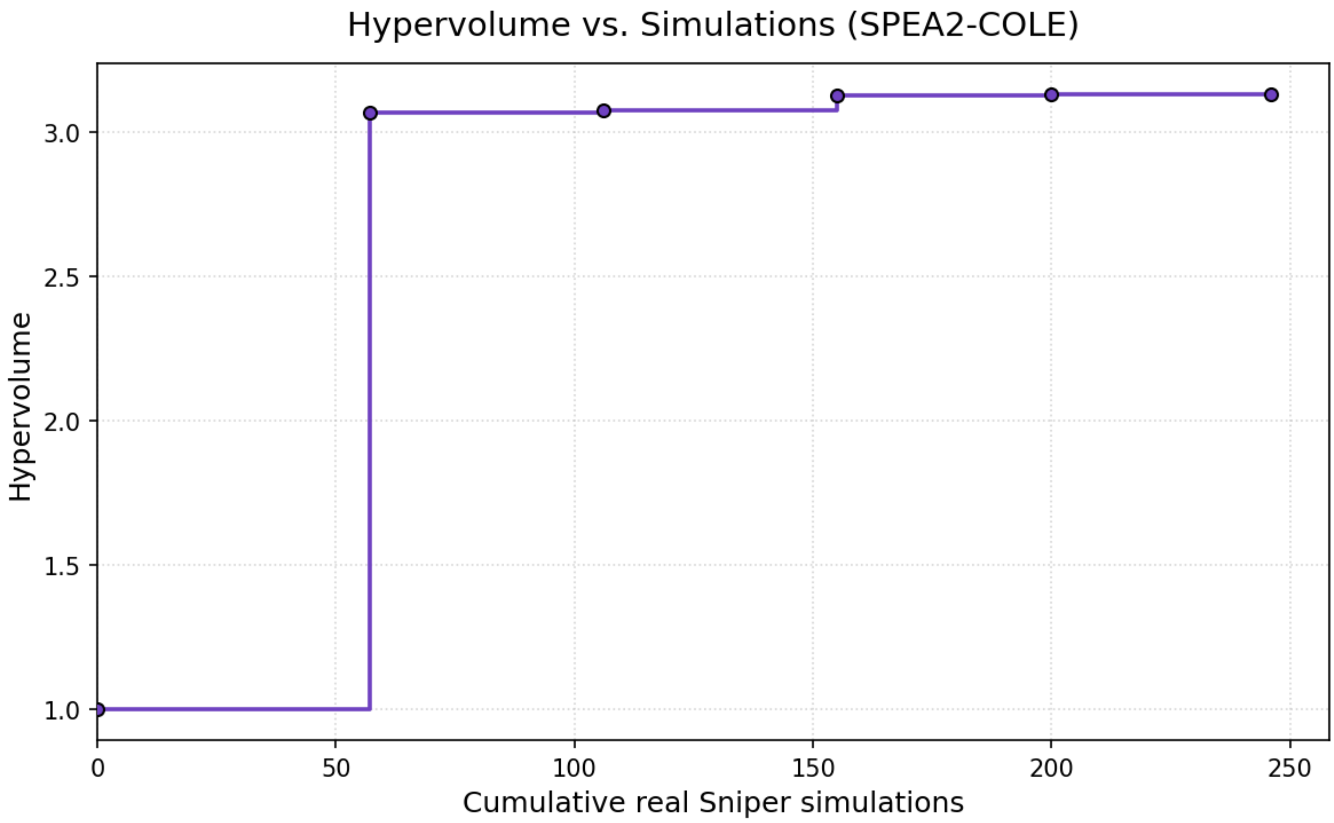
Became infeasable with new parameterspace. first iteration: 33 simulations second iteration: >100 simulations

SPEA2 - COLE

hyperparameter settings 1

Hyperparameter settings: Patience = 1 (amount of iterations to wait before stopping after no improvement), max_iterations: int = 30, folowing parameters are default: num_populations: int = 3, population_size: int = 20, archive_size: int = 10, p_mutation: float = 0.10, p_crossover: float = 0.90, p_migration: float = 0.10.





Result:

Total sniper runs: 246 Final hypervolume: 3.1294

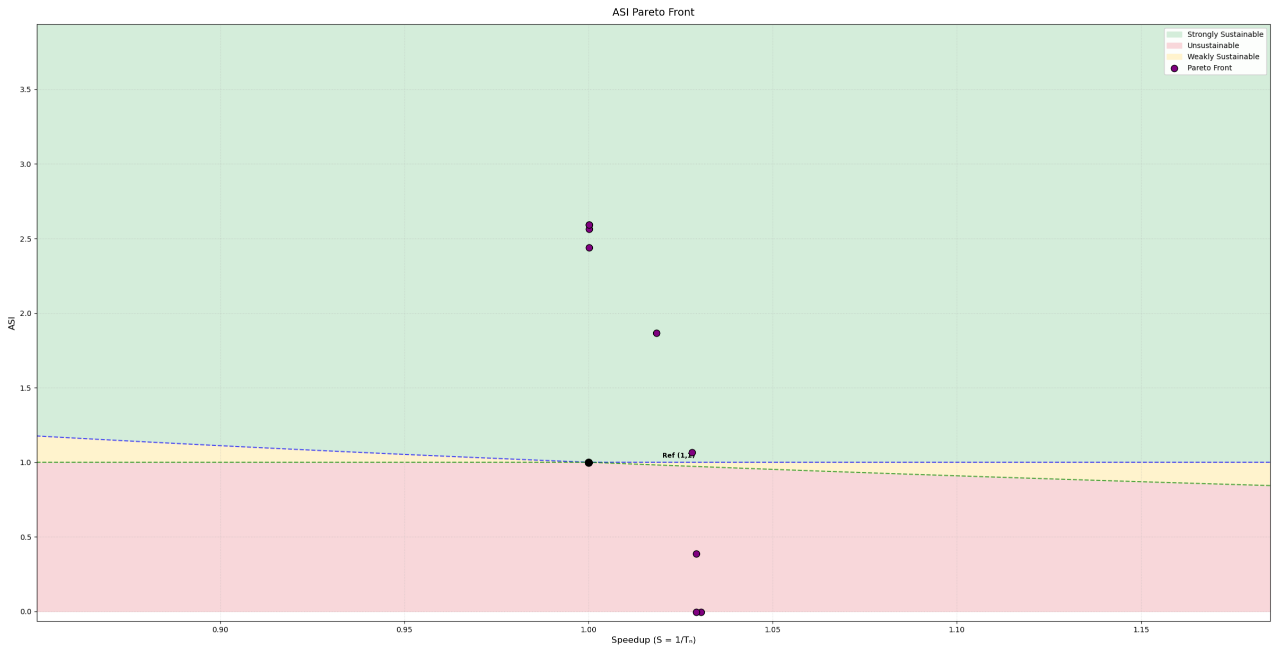
Front:

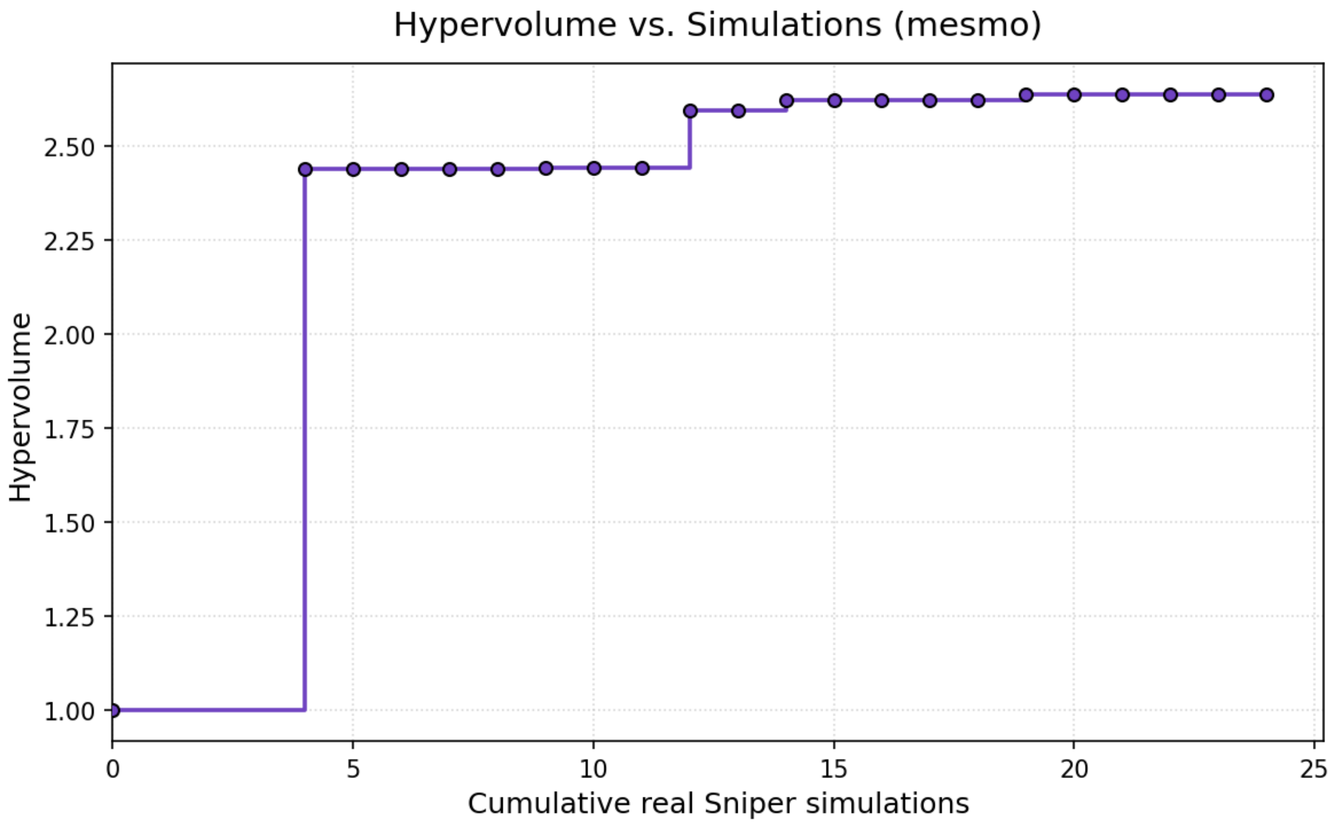
bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
a53	1024	4	64	8	64	8	512	8	2048	3			4	2	1024	1.0624	1.0304	53.11	36.02	Strongly Sust.
a53	1024	4	64	8	64	4	512	8	1024	3			4	2	1024	1.1281	1.0303	47.82	35.26	Strongly Sust.
pentium_m		4	64	4	16	4	512	4	2048				2	2	512	1.5617	1.0302	49.53	25.16	Strongly Sust.
pentium_m		4	64	8	64	8	512	16	4096				10	2	256	1.8130	1.0302	62.43	19.65	Strongly Sust.
pentium_m		8	64	4	64	8	512	16	2048				4	2	256	2.1272	1.0302	53.37	17.96	Strongly Sust.
pentium_m		4	64	4	64	8	512	4	2048				2	2	256	2.2773	1.0302	48.96	17.32	Strongly Sust.
pentium_m		8	64	4	32	4	512	4	1024				2	2	256	2.3633	1.0301	45.75	17.08	Strongly Sust.
pentium_m		8	32	4	32	8	512	4	1024				4	2	256	2.3675	1.0293	45.51	17.08	Strongly Sust.
pentium_m		4	32	4	16	4	512	4	1024				2	2	256	2.5419	1.0293	40.92	16.42	Strongly Sust.
pentium_m		4	32	4	32	8	512	8	1024				4	2	16	2.6494	1.0158	40.92	15.75	Strongly Sust.
a53	512	4	64	4	16	4	256	4	1024	2			4	2	256	2.8179	1.0009	34.41	15.47	Strongly Sust.
pentium_m		4	32	4	16	4	256	4	1024				4	2	256	2.8307	1.0005	34.23	15.41	Strongly Sust.
pentium_m		4	32	4	16	4	256	4	1024				4	2	256	2.8307	1.0005	34.23	15.41	Strongly Sust.
pentium_m		4	32	4	32	8	128	8	1024				4	2	256	2.8878	0.9868	33.52	15.18	Strongly Sust.
a53	512	4	32	4	32	8	128	8	1024	3			4	2	128	2.9270	0.9865	33.40	14.99	Strongly Sust.
a53	1024	4	16	4	32	8	128	8	1024	3			4	2	128	2.9384	0.9861	33.21	14.95	Strongly Sust.
pentium_m		4	16	4	32	8	128	8	1024				4	2	16	3.0553	0.9734	33.00	14.39	Strongly Sust.

MESMO

hyperparameter settings 1

hyperparameter settings: 200 candidate samples (200 candidate configurations are considered per iteration; each is scored by the information gained, averaged over a default of 10 Monte Carlo posterior-function samples (amount is hyperparameter, default is 10), and the top-scoring one (also hyperparameter, default is 1) is evaluated.) are used and 5 initial configurations (including baseline, always the case!) have been run. 20 iterations.





Result:

Total sniper runs: 24 Final hypervolume: 2.6384

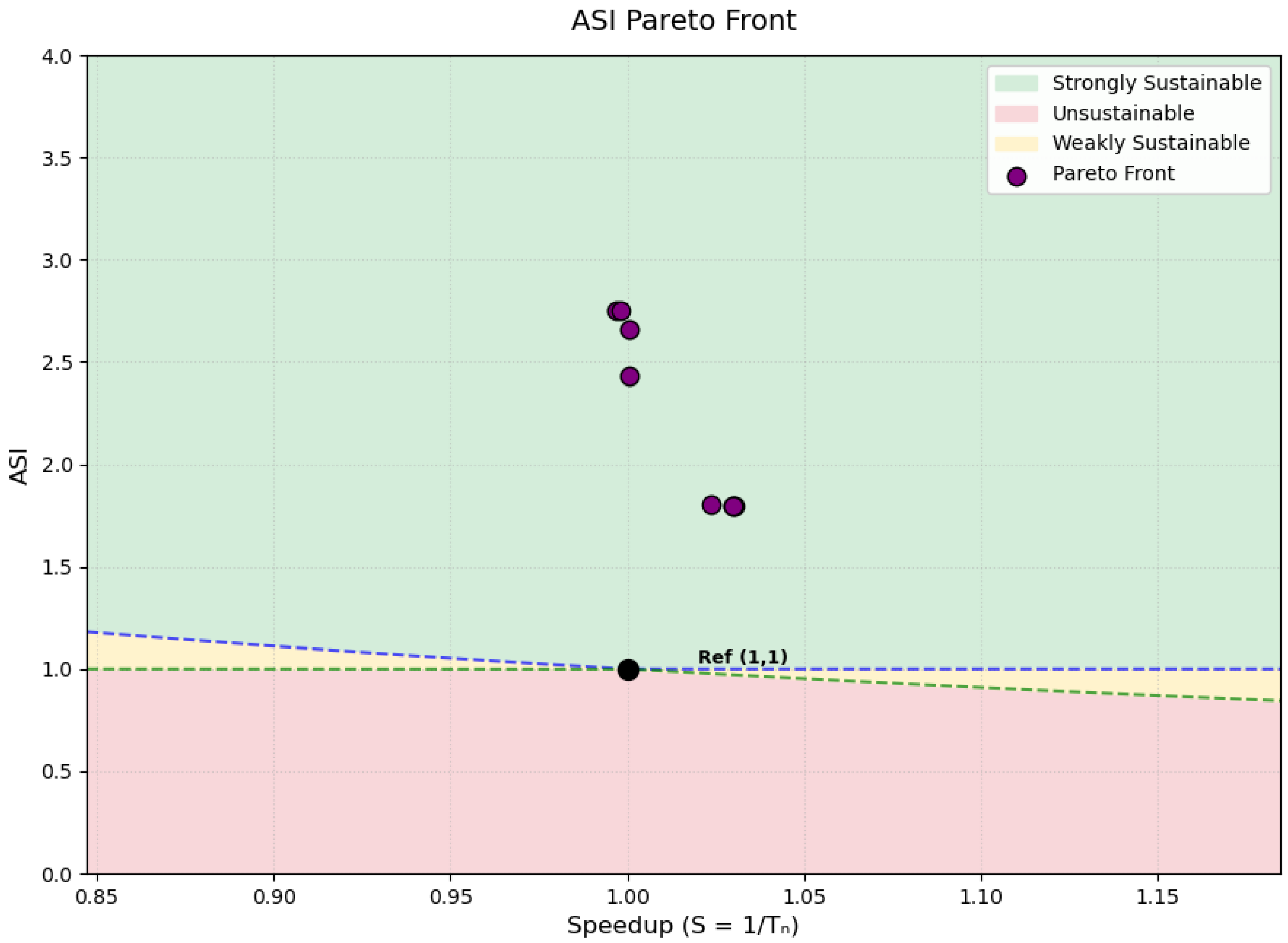
Front:

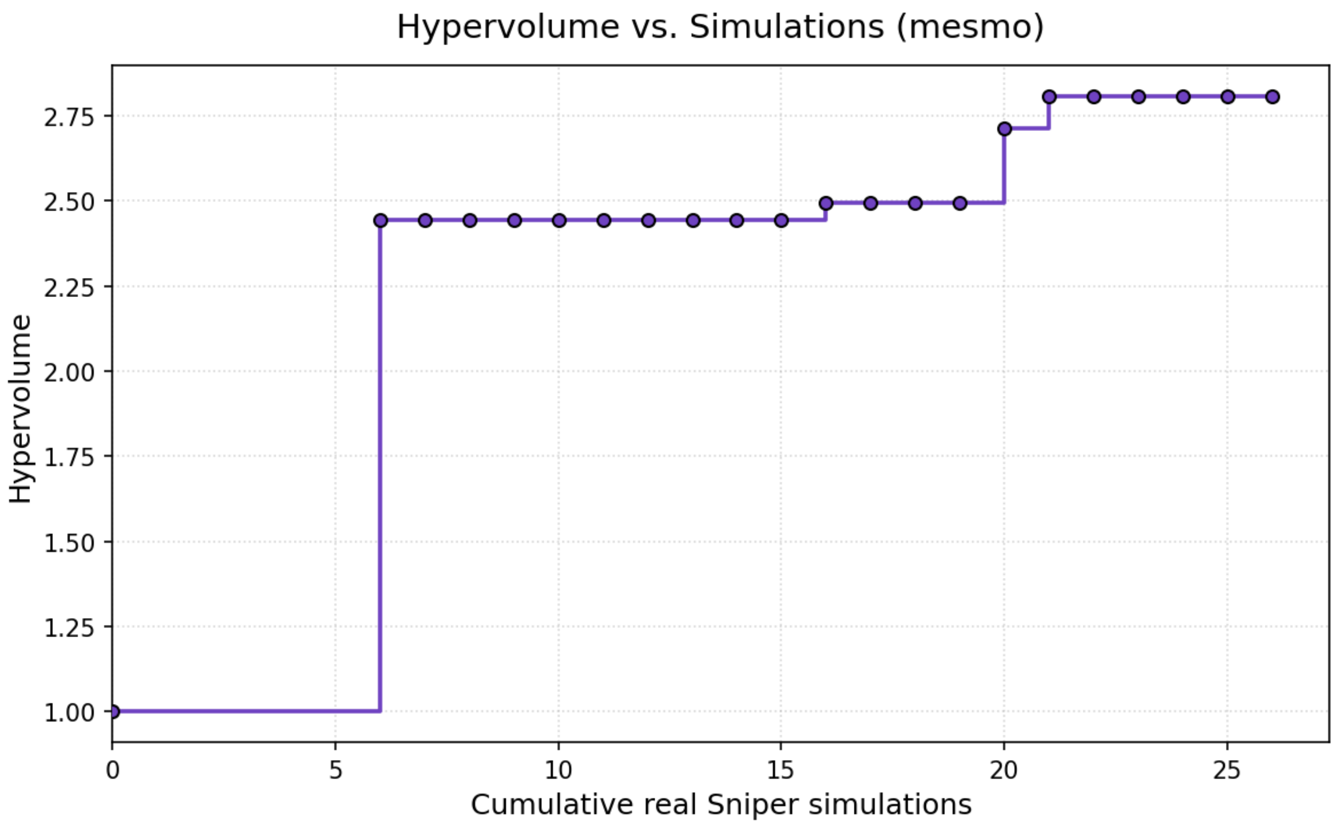
bpt	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m	8	64	4	16	8	512	4	1024			8	10	1024	-0.0056	1.0305	239.58	2631.93	Unsustainable
pentium_m	8	16	4	16	8	512	4	1024			8	10	1024	-0.0054	1.0293	238.13	2631.81	Unsustainable
tage	4	32	4	64	8	512	8	2048			8	4	512	0.3855	1.0293	57.08	96.36	Unsustainable
nn	8	16	4	16	8	512	8	2048	32	0.0005	2	2	1024	1.0671	1.0281	53.81	35.68	Strongly Sust.
tage	4	32	4	64	8	512	8	2048			8	4	16	1.8672	1.0185	49.44	21.06	Strongly Sust.
pentium_m	4	16	8	16	4	256	4	2048			2	2	256	2.4405	1.0001	42.61	16.90	Strongly Sust.
pentium_m	4	16	4	64	4	256	4	2048			2	2	256	2.5637	1.0001	41.82	16.18	Strongly Sust.
pentium_m	4	16	4	16	4	256	4	2048			8	2	256	2.5931	1.0001	40.99	16.09	Strongly Sust.
pentium_m	4	16	4	16	4	256	4	2048			2	2	256	2.5931	1.0001	40.99	16.09	Strongly Sust.

hyperparameter settings 2

hyperparameter settings: 400 candidate samples are used and 7 initial configurations have been run.

20 iterations





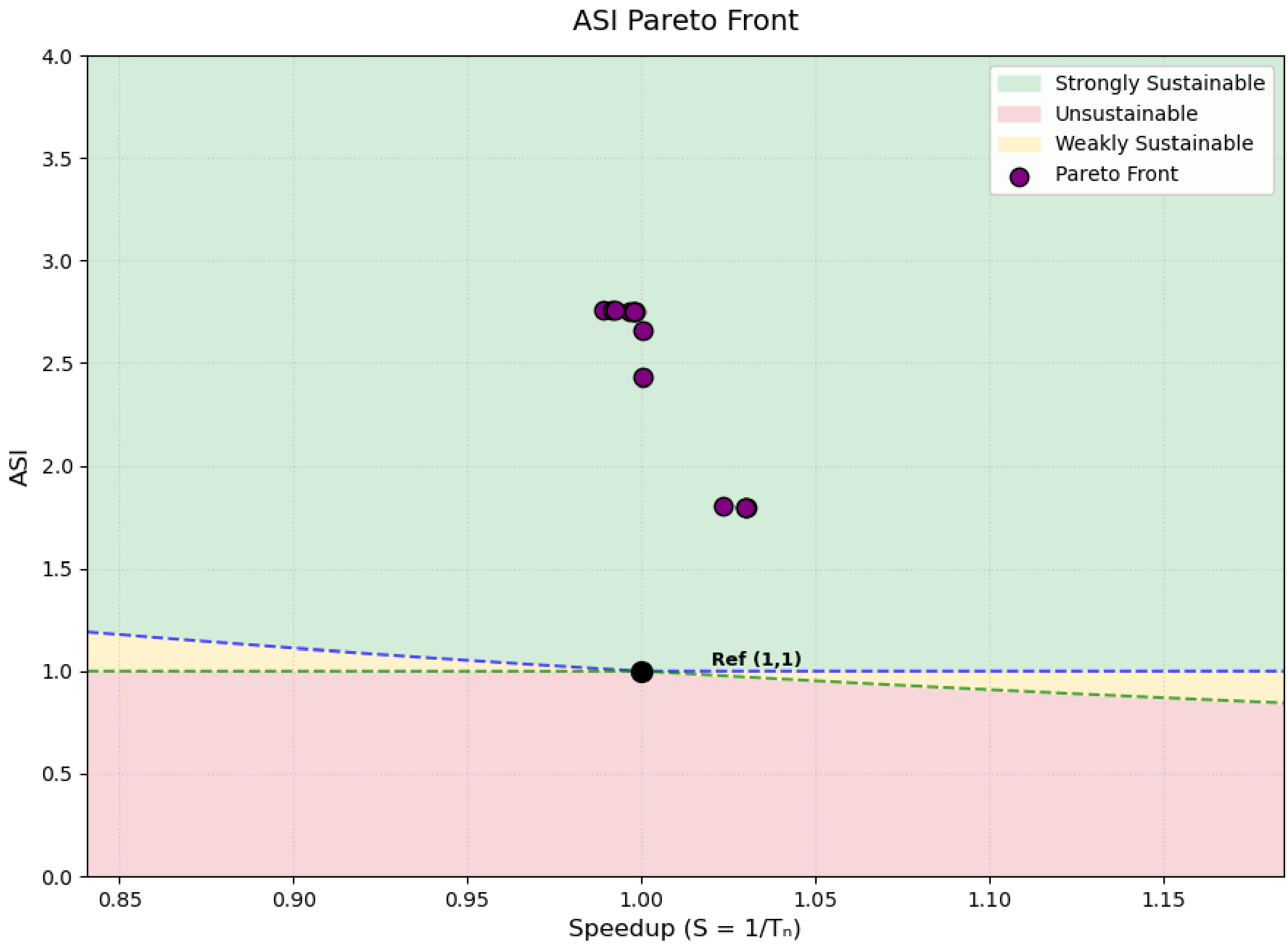
Result:

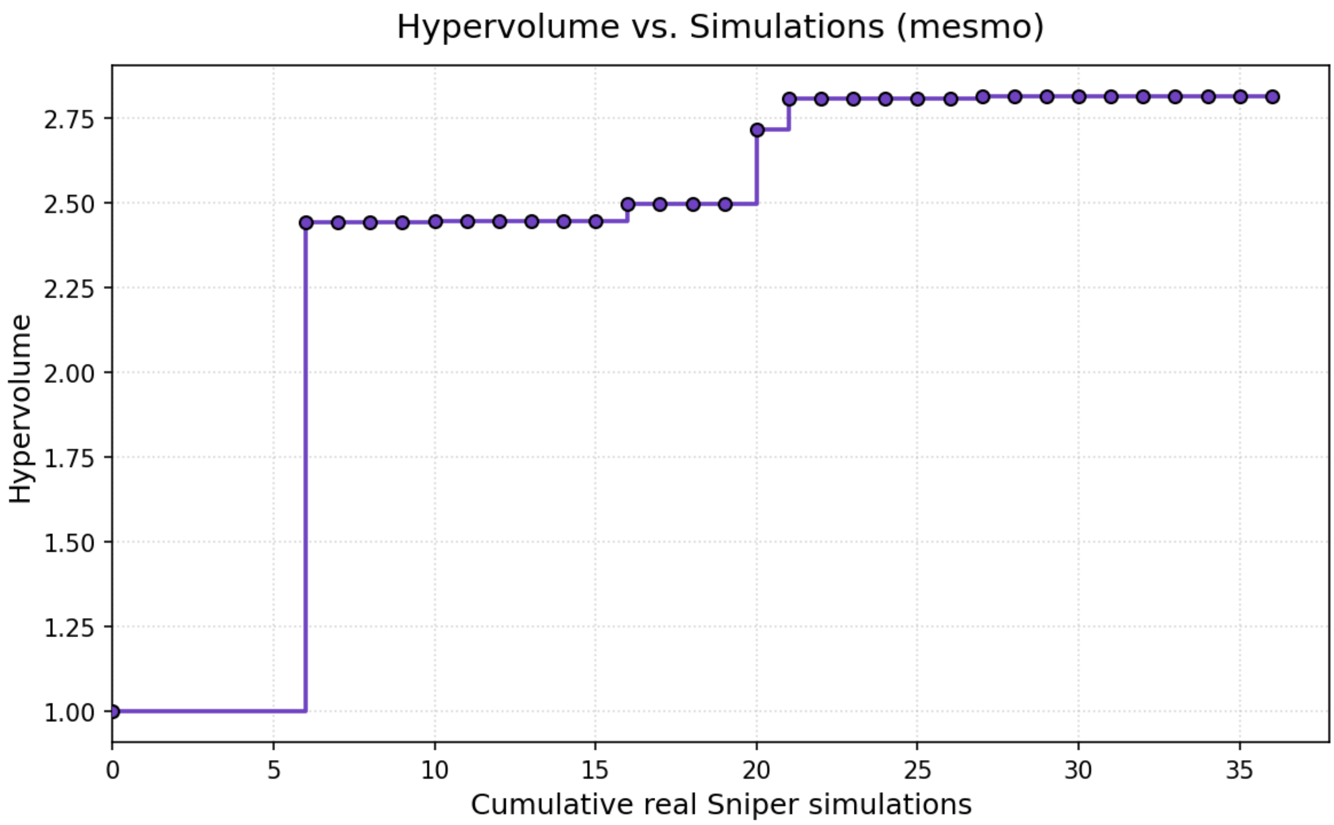
Total sniper runs: 26 Final hypervolume: 2.8076

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m		8	64	4	32	4	512	4	4096				2	2	256	1.7997	1.0301	65.13	19.37	Strongly Sust.
a53	512	8	64	4	32	4	512	4	4096	4			2	2	256	1.7998	1.0298	65.13	19.37	Strongly Sust.
a53	512	8	64	4	32	4	512	4	4096	4			10	2	256	1.7998	1.0298	65.13	19.37	Strongly Sust.
nn		8	64	4	32	4	512	4	4096		16	0.001	2	2	256	1.8013	1.0235	65.13	19.35	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	2048				2	2	256	2.4315	1.0005	42.80	16.94	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	1024				2	2	256	2.6597	1.0005	35.85	16.23	Strongly Sust.
nn		4	32	8	16	4	256	4	1024		16	0.005	2	2	32	2.7519	0.9982	35.61	15.71	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	1024				2	2	32	2.7531	0.9967	35.61	15.71	Strongly Sust.

30 iterations





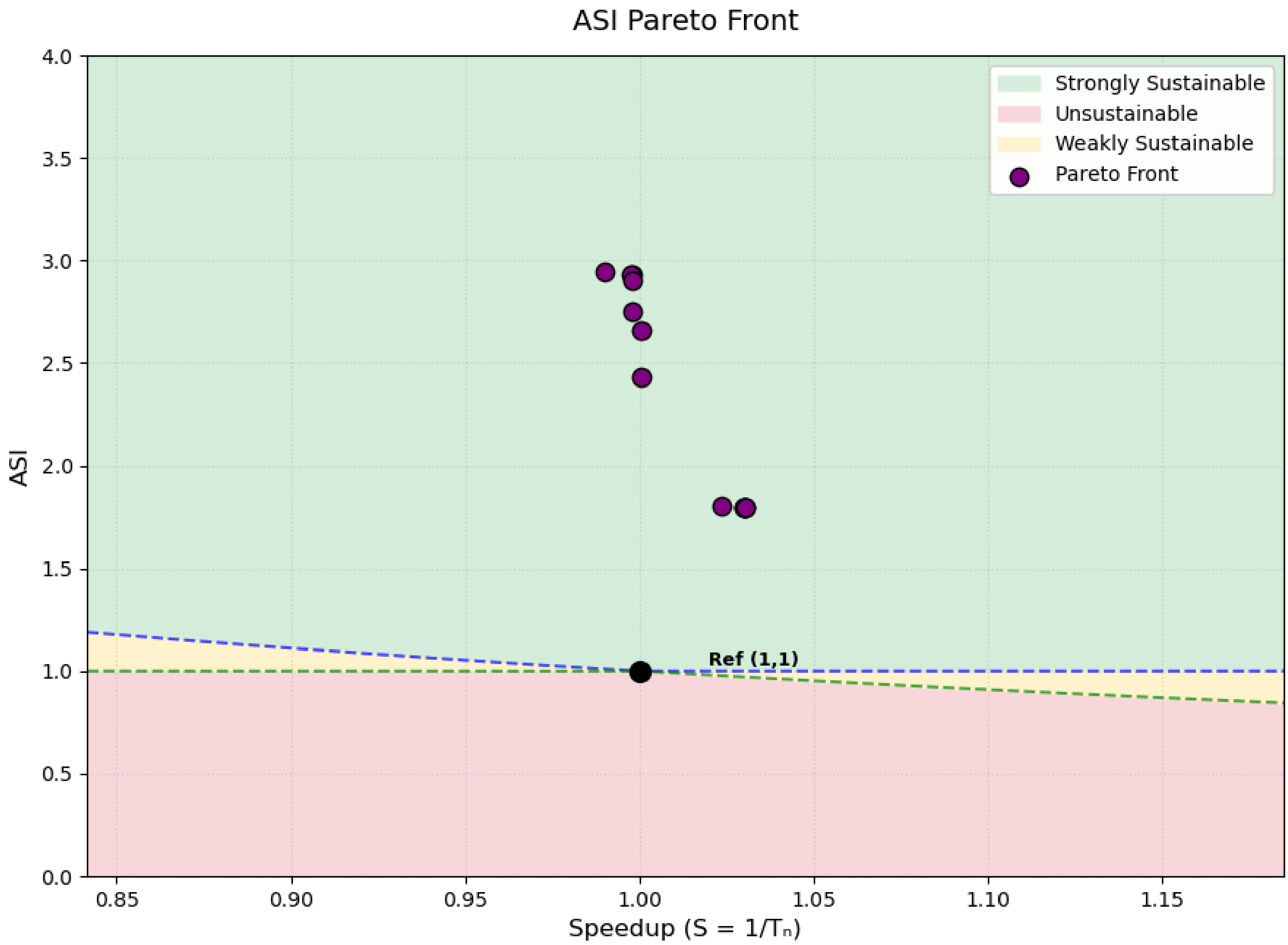
Result:

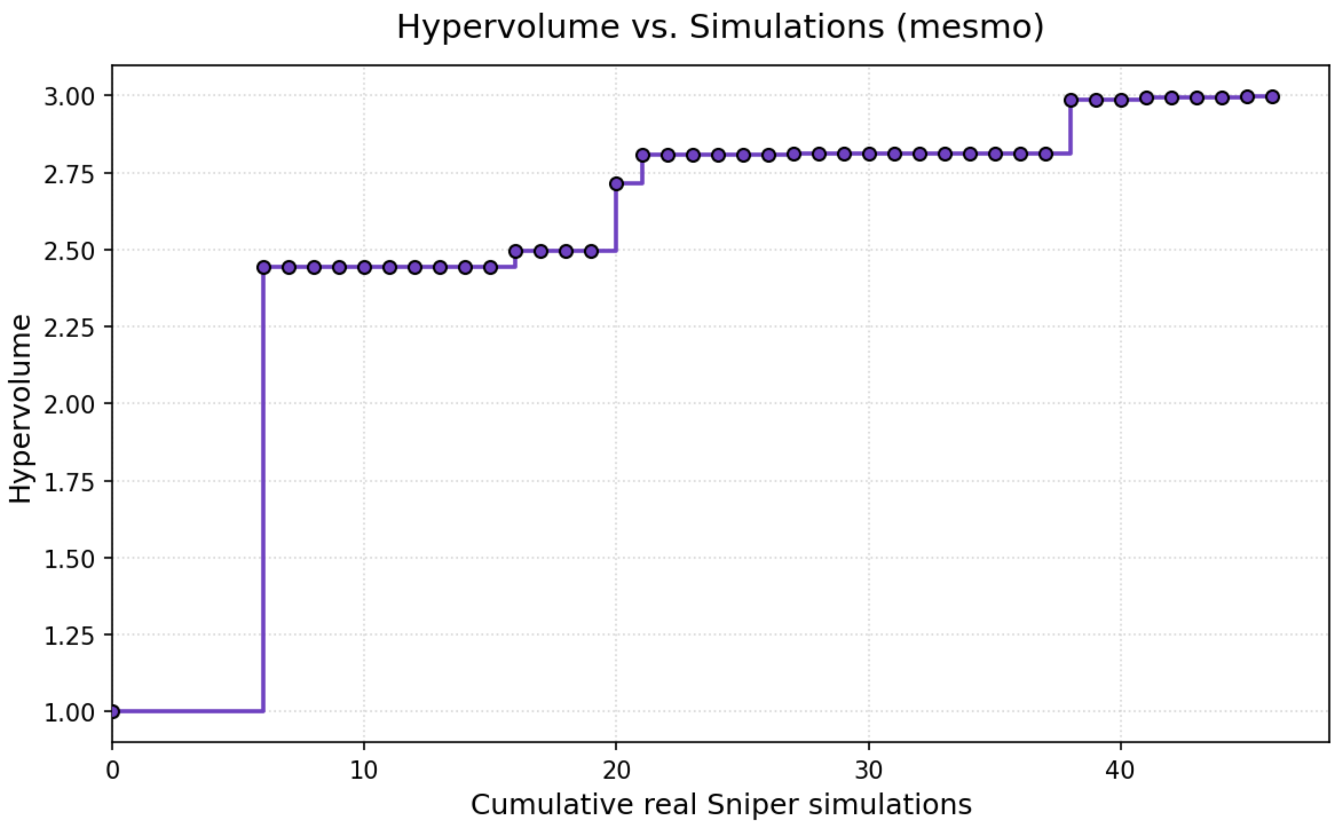
Total sniper runs: 36 Final hypervolume: 2.8132

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m		8	64	4	32	4	512	4	4096				2	2	256	1.7997	1.0301	65.13	19.37	Strongly Sust.
a53	512	8	64	4	32	4	512	4	4096	4			2	2	256	1.7998	1.0298	65.13	19.37	Strongly Sust.
a53	512	8	64	4	32	4	512	4	4096	4			10	2	256	1.7998	1.0298	65.13	19.37	Strongly Sust.
nn		8	64	4	32	4	512	4	4096		16	0.001	2	2	256	1.8013	1.0235	65.13	19.35	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	2048				2	2	256	2.4315	1.0005	42.80	16.94	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	1024				2	2	256	2.6597	1.0005	35.85	16.23	Strongly Sust.
nn		4	32	8	16	4	256	4	1024		16	0.005	2	2	32	2.7519	0.9982	35.61	15.71	Strongly Sust.
a53	2048	4	32	8	16	4	256	4	1024	4			2	2	32	2.7520	0.9980	35.61	15.71	Strongly Sust.
tage		4	32	8	16	4	256	4	1024				2	2	32	2.7522	0.9978	35.61	15.71	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	1024				2	2	32	2.7531	0.9967	35.61	15.71	Strongly Sust.
a53	1024	4	32	8	16	4	256	4	1024	2			2	2	32	2.7567	0.9923	35.61	15.68	Strongly Sust.
nn		4	32	8	16	4	256	4	1024		64	0.0005	2	2	32	2.7573	0.9915	35.61	15.68	Strongly Sust.
nn		4	32	8	16	4	256	4	1024		64	0.005	2	2	32	2.7587	0.9892	35.61	15.67	Strongly Sust.

40 iterations





Result:

Total sniper runs: 46 Final hypervolume: 2.9982

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m		8	64	4	32	4	512	4	4096				8	2	256	1.7997	1.0302	65.13	19.37	Strongly Sust.
pentium_m		8	64	4	32	4	512	4	4096				2	2	256	1.7997	1.0301	65.13	19.37	Strongly Sust.
a53	512	8	64	4	32	4	512	4	4096	4			2	2	256	1.7998	1.0298	65.13	19.37	Strongly Sust.
a53	512	8	64	4	32	4	512	4	4096	4			10	2	256	1.7998	1.0298	65.13	19.37	Strongly Sust.
nn		8	64	4	32	4	512	4	4096		16	0.001	2	2	256	1.8013	1.0235	65.13	19.35	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	2048				2	2	256	2.4315	1.0005	42.80	16.94	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	1024				2	2	256	2.6597	1.0005	35.85	16.23	Strongly Sust.
nn		4	32	8	16	4	256	4	1024		16	0.005	2	2	32	2.7519	0.9982	35.61	15.71	Strongly Sust.
a53	2048	4	32	4	64	4	256	4	1024	4			2	2	32	2.8998	0.9981	34.82	14.99	Strongly Sust.
a53	2048	4	32	4	16	4	256	4	1024	4			2	2	32	2.9337	0.9980	34.00	14.89	Strongly Sust.
a53	1024	4	32	4	16	4	256	4	1024	4			2	2	32	2.9341	0.9975	34.00	14.89	Strongly Sust.
a53	2048	4	32	4	64	4	256	4	1024	4			2	2	16	2.9440	0.9902	34.74	14.77	Strongly Sust.

50 iterations

Mistake with automaticly plotting, but conclusion is that hypervolume keeps increasing, also for 60 iters it will even be higher (3.0375)

Result:

Total sniper runs: 56 Final hypervolume: 3.0285

Front:

50 iters:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
tage		8	64	8	16	4	512	4	4096				8	2	1024	0.8705	1.0304	70.96	38.15	Unsustainable
pentium_m		8	64	8	32	8	512	16	2048				10	2	512	1.4025	1.0302	56.00	26.71	Strongly Sust.
pentium_m		8	64	8	32	8	512	16	2048				8	2	512	1.4025	1.0302	56.00	26.71	Strongly Sust.
pentium_m		8	64	4	32	4	512	4	4096				8	2	256	1.7997	1.0302	65.13	19.37	Strongly Sust.
pentium_m		8	64	4	32	4	512	4	4096				2	2	256	1.7997	1.0301	65.13	19.37	Strongly Sust.
pentium_m		8	64	8	32	8	512	16	2048				8	2	128	2.0450	1.0299	54.39	18.54	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	2048				2	2	256	2.4315	1.0005	42.80	16.94	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	1024				2	2	256	2.6597	1.0005	35.85	16.23	Strongly Sust.
nn		4	32	8	16	4	256	4	1024		16	0.005	2	2	32	2.7519	0.9982	35.61	15.71	Strongly Sust.
a53	2048	4	32	4	64	4	256	4	1024	4			2	2	32	2.8998	0.9981	34.82	14.99	Strongly Sust.
a53	2048	4	32	4	16	4	256	4	1024	4			2	2	32	2.9337	0.9980	34.00	14.89	Strongly Sust.
a53	1024	4	32	4	16	4	256	4	1024	4			2	2	32	2.9341	0.9975	34.00	14.89	Strongly Sust.
a53	2048	4	32	4	64	4	256	4	1024	4			2	2	16	2.9440	0.9902	34.74	14.77	Strongly Sust.
pentium_m		4	32	4	64	4	256	4	1024				2	2	16	2.9461	0.9874	34.74	14.76	Strongly Sust.
nn		4	32	4	64	4	256	4	1024		64	0.001	2	2	16	2.9679	0.9577	34.74	14.65	Strongly Sust.

60 iters:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
tage		8	64	8	16	4	512	4	4096				8	2	1024	0.8705	1.0304	70.96	38.15	Unsustainable
pentium_m		8	64	8	32	8	512	16	2048				10	2	512	1.4025	1.0302	56.00	26.71	Strongly Sust.
pentium_m		8	64	8	32	8	512	16	2048				8	2	512	1.4025	1.0302	56.00	26.71	Strongly Sust.
pentium_m		8	64	4	32	4	512	4	4096				8	2	256	1.7997	1.0302	65.13	19.37	Strongly Sust.
pentium_m		8	64	4	32	4	512	4	4096				2	2	256	1.7997	1.0301	65.13	19.37	Strongly Sust.
pentium_m		8	64	8	32	8	512	16	2048				8	2	128	2.0450	1.0299	54.39	18.54	Strongly Sust.
a53	2048	8	32	4	64	4	512	16	1024	4			8	2	64	2.3621	1.0287	47.90	16.83	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	2048				2	2	256	2.4315	1.0005	42.80	16.94	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	1024				2	2	256	2.6597	1.0005	35.85	16.23	Strongly Sust.
nn		4	32	8	16	4	256	4	1024		16	0.005	2	2	32	2.7519	0.9982	35.61	15.71	Strongly Sust.
a53	2048	4	32	4	64	4	256	4	1024	4			2	2	32	2.8998	0.9981	34.82	14.99	Strongly Sust.
a53	2048	4	32	4	16	4	256	4	1024	4			2	2	32	2.9337	0.9980	34.00	14.89	Strongly Sust.
a53	1024	4	32	4	16	4	256	4	1024	4			2	2	32	2.9341	0.9975	34.00	14.89	Strongly Sust.
a53	2048	4	32	4	64	4	256	4	1024	4			2	2	16	2.9440	0.9902	34.74	14.77	Strongly Sust.
pentium_m		4	32	4	64	4	256	4	1024				8	2	16	2.9461	0.9874	34.74	14.76	Strongly Sust.
pentium_m		4	32	4	64	4	256	4	1024				2	2	16	2.9461	0.9874	34.74	14.76	Strongly Sust.
a53	512	4	32	4	64	4	256	4	1024	4			2	2	16	2.9463	0.9871	34.74	14.76	Strongly Sust.
nn		4	32	4	64	4	256	4	1024		16	0.001	10	2	16	2.9521	0.9789	34.74	14.73	Strongly Sust.
nn		4	32	4	64	4	256	4	1024		16	0.001	2	2	16	2.9678	0.9580	34.74	14.65	Strongly Sust.
nn		4	32	4	64	4	256	4	1024		64	0.001	2	2	16	2.9679	0.9577	34.74	14.65	Strongly Sust.

ML2, CCl, MIP and EI benchmarks:

ML2: L2 resident (depending on LFSR settings) linked list traversal

CCl: Impossible control with large basic blocks (potentially larger penalty)

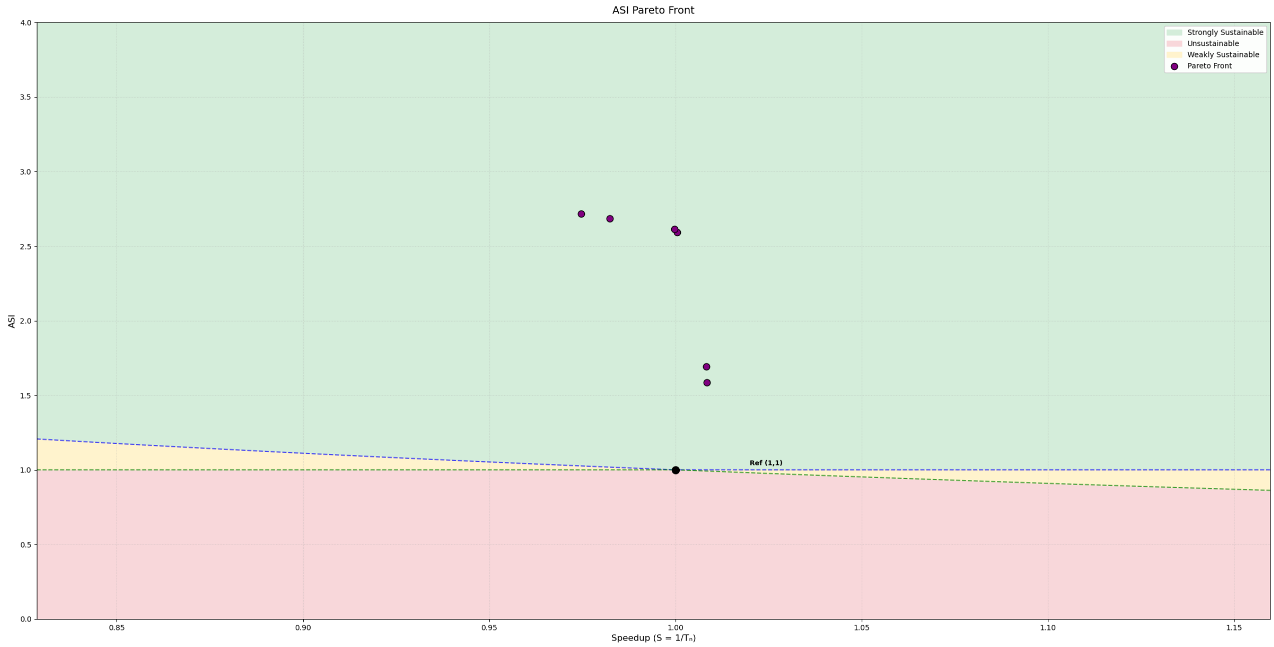
MIP: Large Instruction Region -- Instruction cache Misses

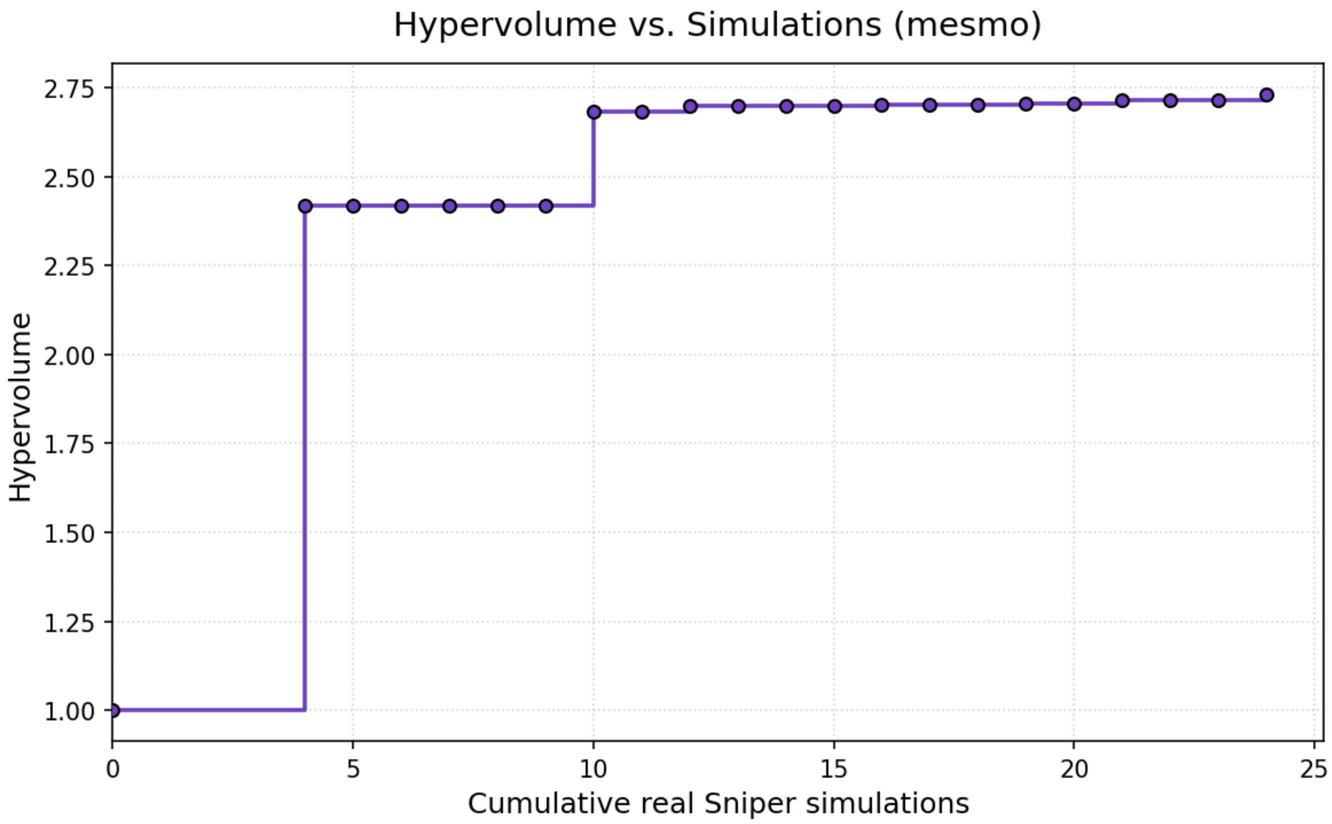
EI: Integer Execution -- 8 Independent computations per iteration

MESMO

hyperparameter settings 1

hyperparameter settings: 200 candidate samples are used and 5 initial configurations are ran. 20 iterations.





Result:

Total sniper runs: 24 Final hypervolume: 2.7310

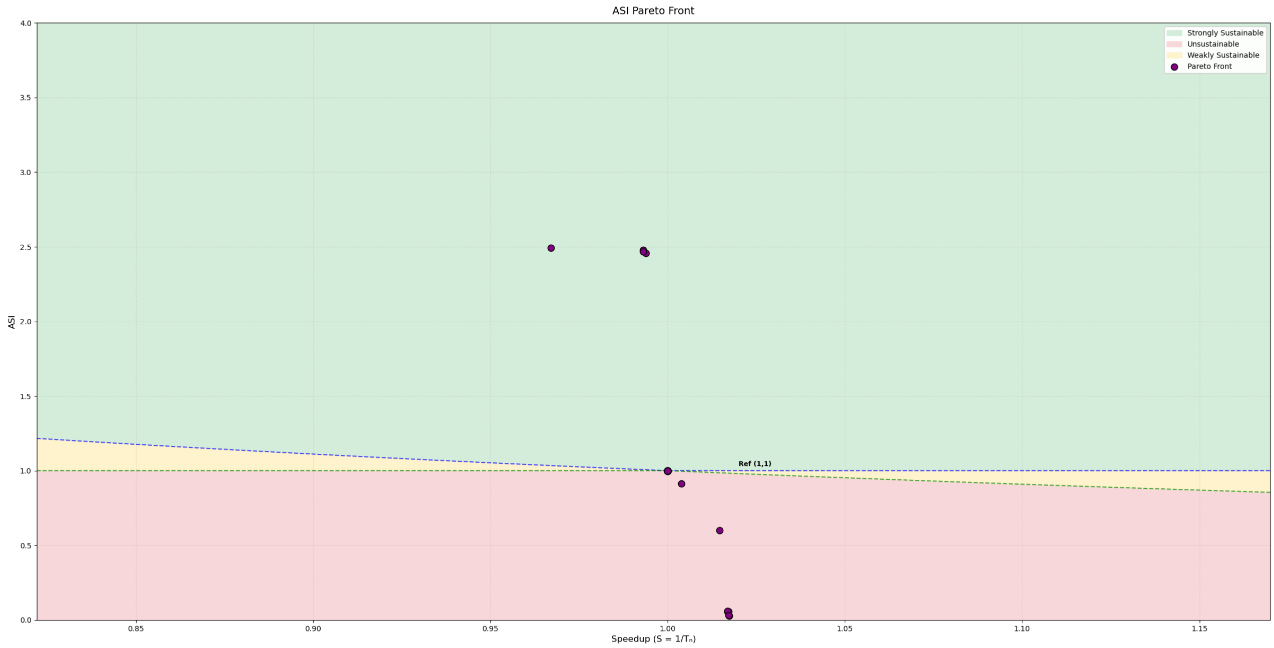
Front:

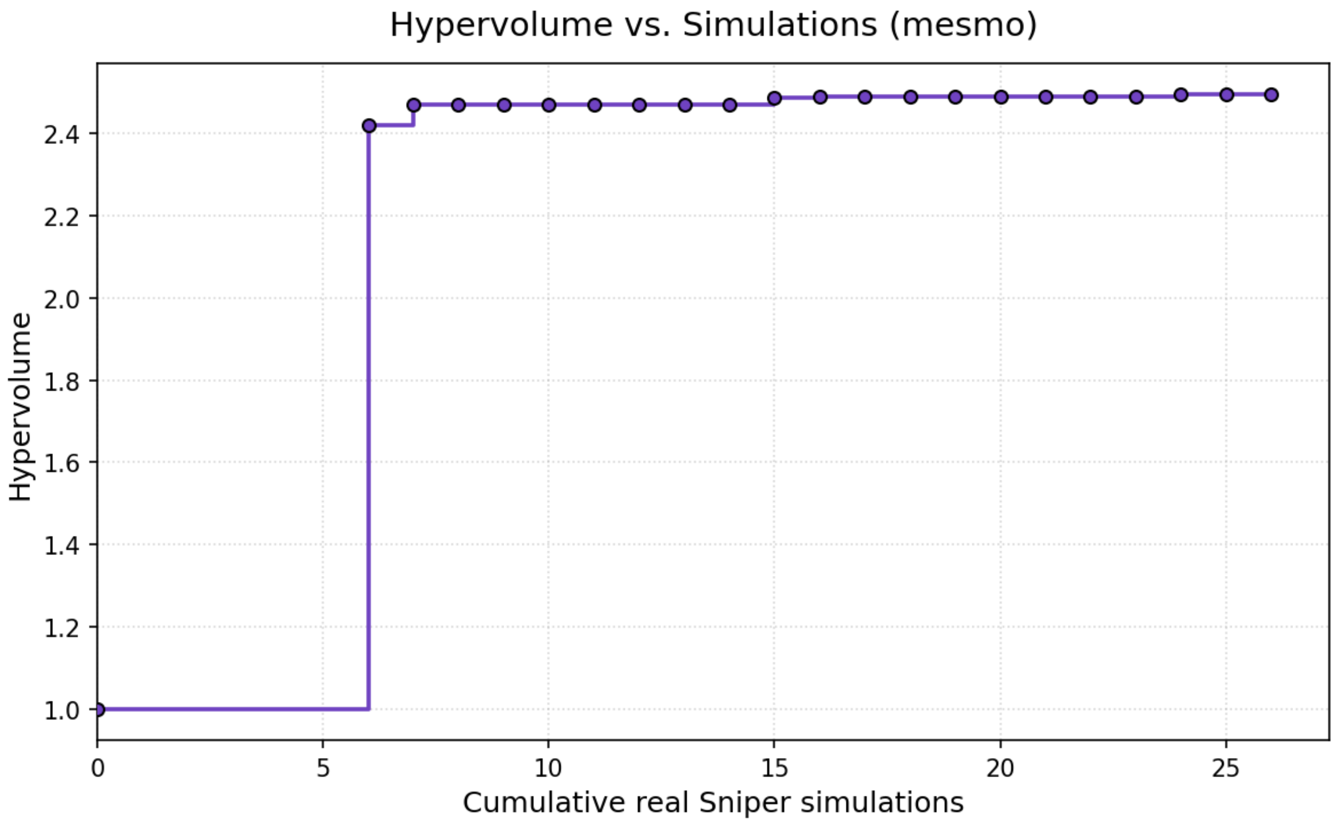
bpt	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m	4	64	4	32	8	512	8	2048			2	2	512	1.5862	1.0085	48.11	25.86	Strongly Sust.
pentium_m	4	64	4	32	8	512	4	1024			2	2	512	1.6928	1.0083	42.91	25.13	Strongly Sust.
pentium_m	4	64	4	32	8	512	4	1024			2	2	32	2.5911	1.0004	41.18	16.62	Strongly Sust.
pentium_m	4	16	4	32	8	512	4	1024			2	2	32	2.6146	0.9998	40.81	16.51	Strongly Sust.

bpt	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
nn	8	16	4	32	8	256	4	1024	64	0.005	2	2	64	2.6838	0.9823	37.99	16.39	Strongly Sust.
nn	8	16	4	32	8	256	4	1024	64	0.005	2	2	32	2.7184	0.9747	37.93	16.19	Strongly Sust.

hyperparameter settings 2

hyperparameter settings: 400 candidate samples are used and 5 initial configurations are ran. 20 iterations. 4 benchmarks: ML2, CCl, MIP and EI.





Result:

Total sniper runs: 26 Final hypervolume: 2.4945

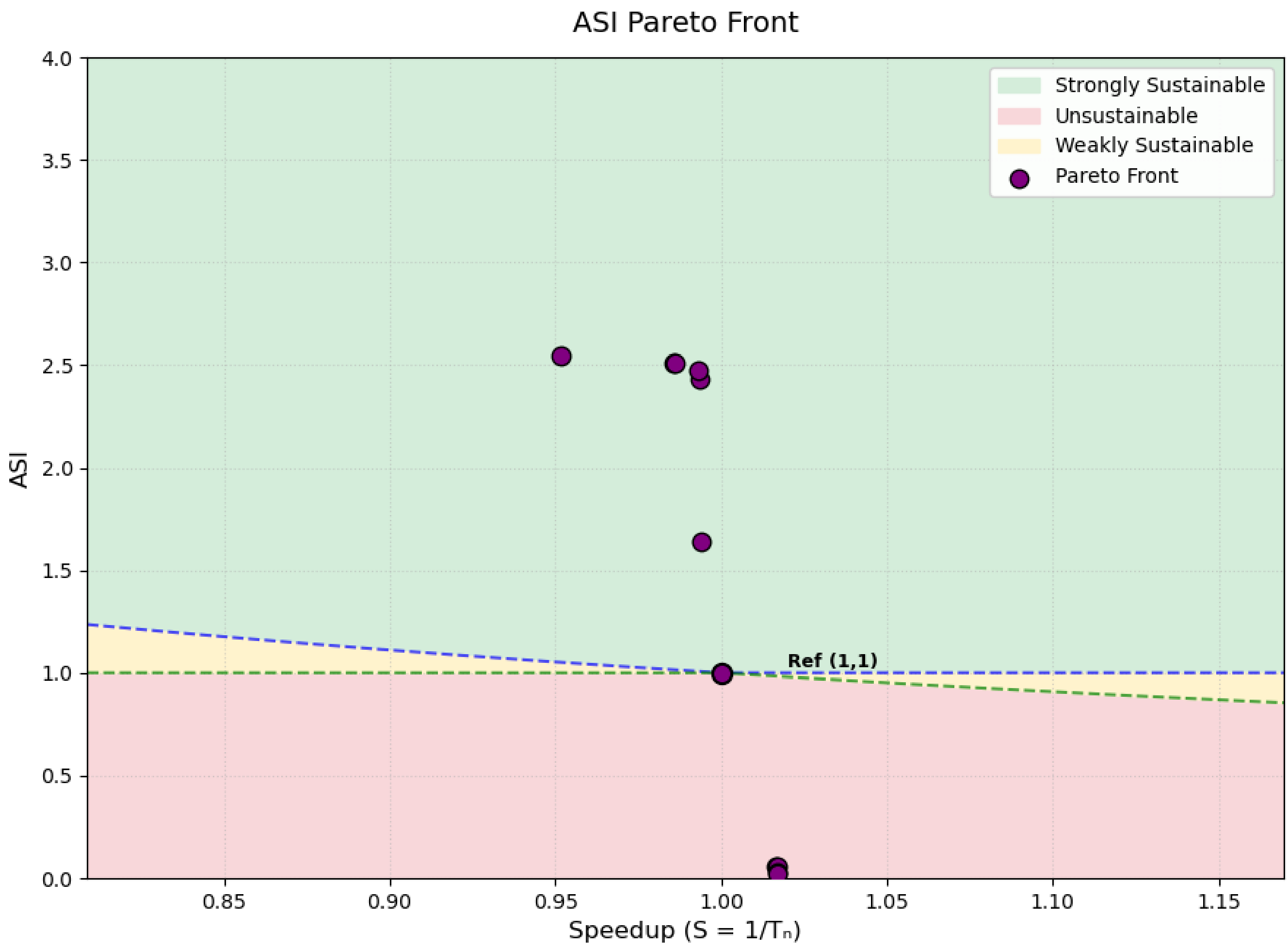
Front:

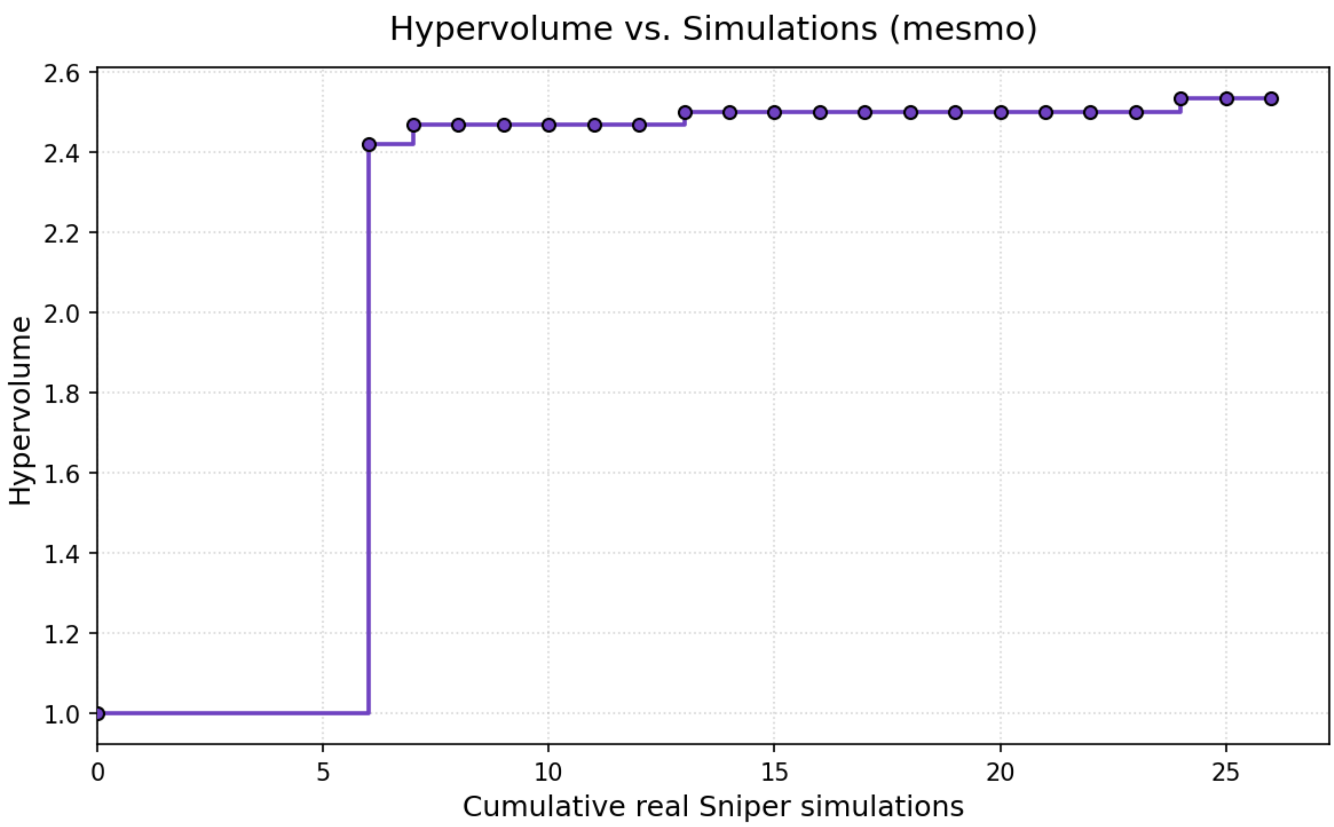
bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m		4	64	4	16	8	512	8	4096				8	8	512	0.0267	1.0174	118.45	763.80	Unsustainable
a53	512	8	64	8	16	8	512	16	1024	2			10	8	512	0.0310	1.0172	107.21	763.19	Unsustainable
pentium_m		8	64	8	16	8	512	16	1024				10	8	256	0.0572	1.0171	83.61	535.26	Unsustainable
a53	512	8	64	8	16	8	512	16	1024	2			10	8	256	0.0572	1.0170	83.61	535.26	Unsustainable
a53	512	4	64	8	16	8	512	16	1024	2			10	8	256	0.0597	1.0170	79.19	534.47	Unsustainable
pentium_m		8	64	4	32	8	512	8	1024				2	10	64	0.6014	1.0147	63.70	60.61	Unsustainable
tage		8	64	4	16	4	512	8	1024				10	8	32	0.9122	1.0039	56.76	42.19	Unsustainable
baseline																1.0000	1.0000	94.01	27.56	Reference
pentium_m		4	64	8	16	4	256	4	2048				2	2	64	2.4560	0.9938	42.80	17.34	Strongly Sust.
pentium_m		4	16	8	32	4	256	4	2048				2	2	64	2.4675	0.9931	42.78	17.26	Strongly Sust.
pentium_m		4	16	8	16	4	256	4	2048				2	2	64	2.4779	0.9931	42.43	17.23	Strongly Sust.
nn		4	16	8	32	4	256	4	2048		64	0.005	2	2	64	2.4940	0.9671	42.78	17.07	Strongly Sust.

hyperparameter settings 3

hyperparameter settings: 400 candidate samples are used and 7 initial configurations are ran.

20 iterations





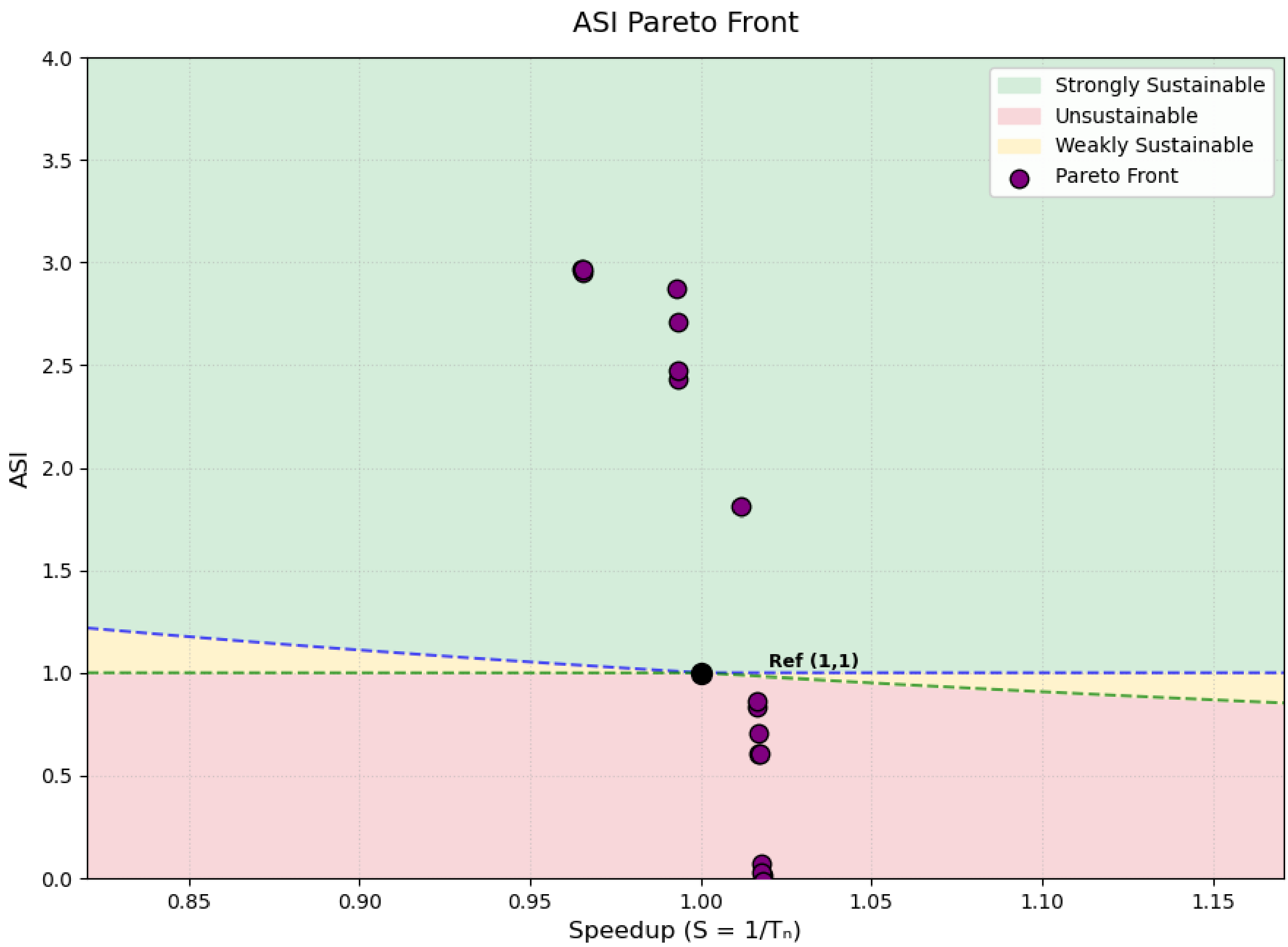
Result:

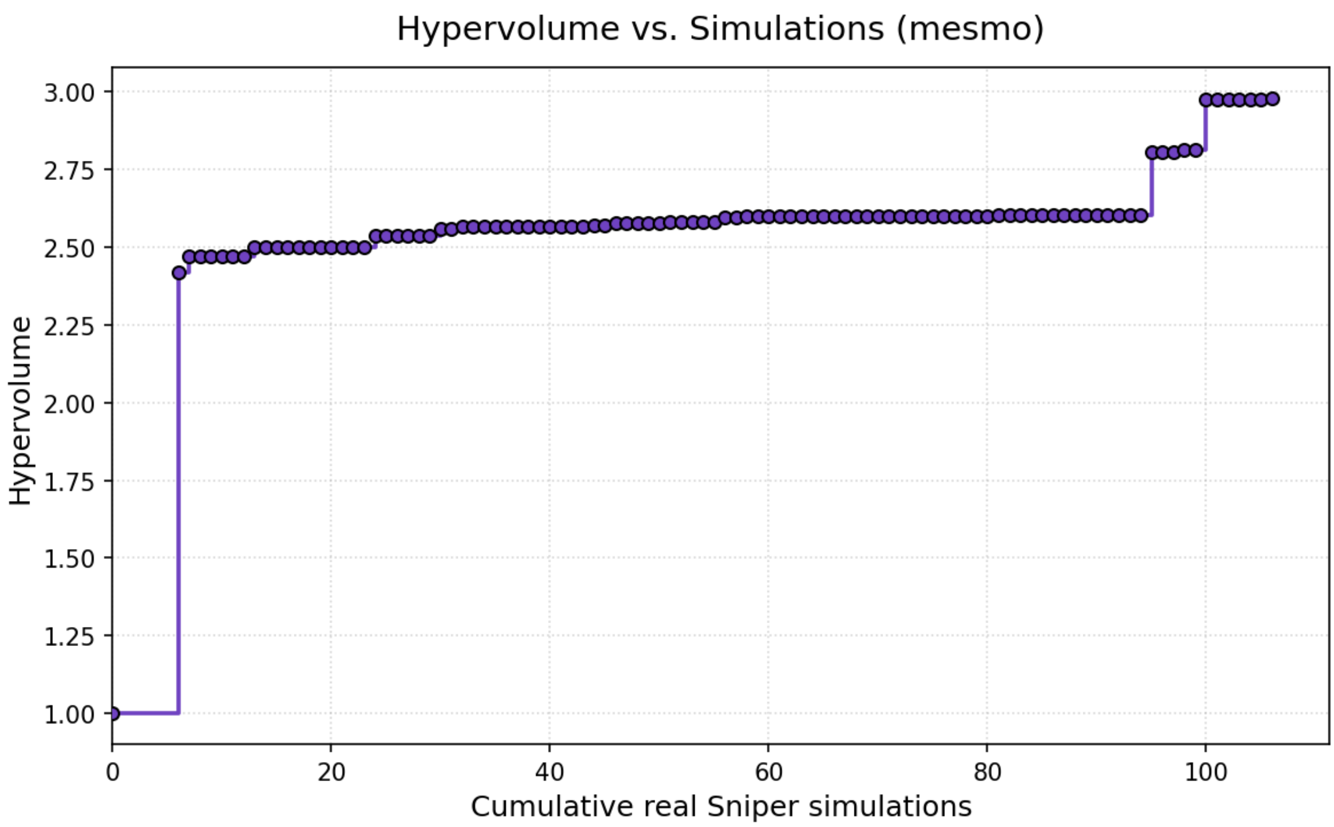
Total sniper runs: 26 Final hypervolume: 2.5351

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
a53	2048	8	32	8	32	4	512	8	2048	4			8	10	256	0.0244	1.0170	105.00	997.11	Unsustainable
pentium_m		8	32	4	16	8	512	8	2048				8	8	512	0.0306	1.0169	108.38	762.95	Unsustainable
a53	1024	8	64	8	16	8	512	8	1024	3			10	8	256	0.0583	1.0169	81.53	535.21	Unsustainable
a53	1024	8	16	8	16	8	512	8	1024	3			10	8	256	0.0591	1.0163	80.08	535.08	Unsustainable
baseline																1.0000	1.0000	94.01	27.56	Reference
pentium_m		4	64	8	16	4	256	4	2048				2	2	512	1.6414	0.9940	44.46	25.64	Strongly Sust.
pentium_m		4	16	8	16	4	256	4	2048				2	2	256	2.4282	0.9933	42.61	17.56	Strongly Sust.
pentium_m		4	16	8	16	4	256	4	2048				2	2	64	2.4777	0.9933	42.43	17.23	Strongly Sust.
tage		4	16	8	16	4	256	4	2048				2	2	32	2.5100	0.9862	42.37	17.01	Strongly Sust.
pentium_m		4	16	8	16	4	256	4	2048				2	2	32	2.5104	0.9857	42.37	17.01	Strongly Sust.
nn		4	16	8	16	4	256	4	2048		64	0.001	2	2	32	2.5457	0.9516	42.37	16.77	Strongly Sust.

100 iterations





Result:

Total sniper runs: 106 Final hypervolume: 2.9794

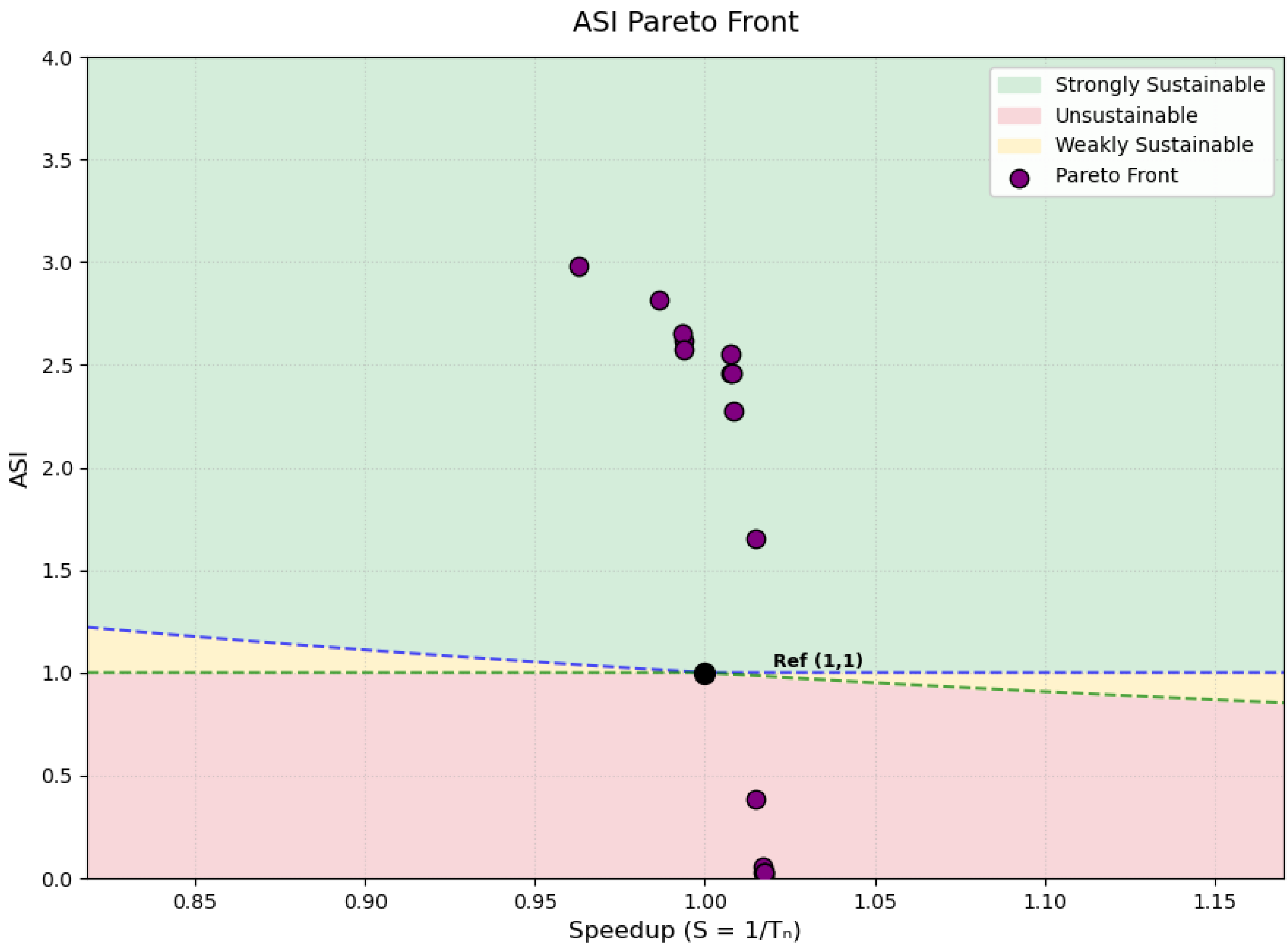
Front:

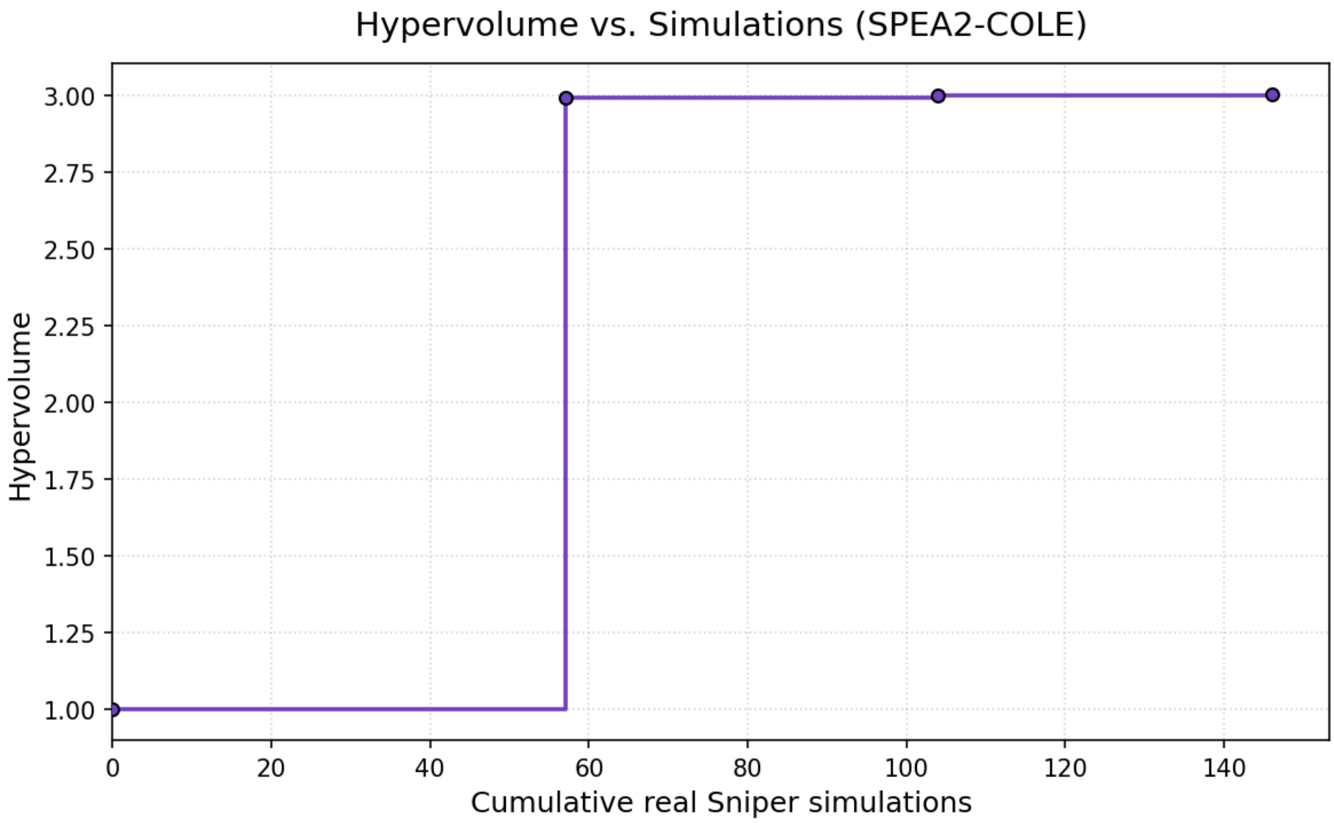
bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
tage		4	64	8	16	4	512	16	8192				8	10	1024	-0.0112	1.0180	288.40	2638.40	Unsustainable
pentium_m		4	64	4	64	8	512	8	1024				10	10	512	0.0115	1.0180	131.82	1434.20	Unsustainable
pentium_m		8	64	4	64	8	512	16	1024				8	10	256	0.0256	1.0177	101.17	995.75	Unsustainable
pentium_m		8	64	4	64	8	512	16	1024				8	10	128	0.0690	1.0175	78.77	464.46	Unsustainable
a53	2048	4	64	8	32	8	512	16	1024	4			8	10	64	0.6029	1.0171	63.08	60.76	Unsustainable
pentium_m		8	32	4	16	8	512	4	1024				8	10	64	0.6048	1.0169	63.13	60.55	Unsustainable
pentium_m		8	64	4	64	8	512	16	4096				8	8	64	0.7077	1.0167	77.22	45.90	Unsustainable
a53	512	4	32	8	64	8	512	4	2048	4			10	8	64	0.8359	1.0163	62.05	44.18	Unsustainable
a53	1024	4	32	4	32	4	512	16	2048	4			10	8	64	0.8658	1.0163	59.92	43.38	Unsustainable
a53	512	8	32	4	16	8	512	8	1024	2			2	4	64	1.8109	1.0115	48.09	22.65	Strongly Sust.
pentium_m		4	16	8	16	4	256	4	2048				2	2	256	2.4282	0.9933	42.61	17.56	Strongly Sust.
pentium_m		4	16	8	16	4	256	4	2048				2	2	64	2.4777	0.9933	42.43	17.23	Strongly Sust.
pentium_m		4	16	8	16	4	256	4	1024				2	2	64	2.7079	0.9931	35.48	16.52	Strongly Sust.
a53	1024	4	32	4	16	4	256	4	1024	2			2	2	64	2.8713	0.9929	34.06	15.72	Strongly Sust.
a53	512	4	64	4	16	4	256	4	1024	2			2	2	16	2.9494	0.9654	34.08	15.30	Strongly Sust.
a53	1024	4	32	4	16	4	256	4	1024	2			2	2	16	2.9640	0.9653	33.91	15.24	Strongly Sust.
a53	512	4	32	4	16	4	256	4	1024	2			2	2	16	2.9642	0.9651	33.91	15.24	Strongly Sust.

SPEA2 - COLE

Hyperparameter settings 1

Hyperparameter settings: Patience = 1 (amount of iterations to wait before stopping after no improvement), max_iterations: int = 10. (max iter never reached)





Result:

Total sniper runs: 146 Final hypervolume: 3.0042

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
a53	2048	4	64	4	32	4	512	16	2048	3			8	10	512	0.0102	1.0176	138.09	1434.83	Unsustainable
a53	2048	4	64	4	32	4	512	16	2048	3			8	10	256	0.0256	1.0174	100.93	995.54	Unsustainable
pentium_m		4	32	4	32	8	512	16	2048				10	10	256	0.0257	1.0174	100.85	995.53	Unsustainable
a53	1024	8	64	8	32	8	512	8	1024	3			4	8	512	0.0317	1.0171	105.48	763.17	Unsustainable
tage		4	32	8	32	4	512	16	4096				10	10	128	0.0603	1.0170	92.12	466.61	Unsustainable
a53	1024	4	64	8	32	8	512	4	2048	3			8	4	512	0.3832	1.0150	60.14	97.85	Unsustainable
tage		8	64	4	32	4	512	8	2048				4	4	128	1.6527	1.0150	53.96	23.78	Strongly Sust.
pentium_m		4	64	4	64	4	512	16	2048				8	2	256	2.2741	1.0084	48.86	17.94	Strongly Sust.
pentium_m		4	64	4	32	8	512	16	1024				8	2	256	2.4611	1.0082	43.49	17.22	Strongly Sust.
pentium_m		4	16	4	64	8	512	16	1024				2	2	256	2.4626	1.0077	43.71	17.18	Strongly Sust.
a53	512	4	32	4	16	8	512	8	1024	3			4	2	128	2.5538	1.0074	40.89	16.89	Strongly Sust.
pentium_m		8	32	4	64	4	256	4	1024				4	2	256	2.5740	0.9936	39.32	16.94	Strongly Sust.
tage		4	64	8	16	8	256	4	1024				8	2	256	2.6172	0.9936	36.56	16.97	Strongly Sust.
pentium_m		4	16	8	16	4	256	4	1024				8	2	256	2.6515	0.9935	35.65	16.85	Strongly Sust.
a53	512	4	32	4	32	4	128	16	1024	3			4	2	256	2.8159	0.9865	35.58	15.87	Strongly Sust.
a53	512	4	16	4	32	8	128	16	1024	3			4	2	16	2.9808	0.9627	35.08	15.04	Strongly Sust.